

# PCA9534A Remote 8-BIT I<sup>2</sup>C and SMBus Low-Power I/O Expander With Interrupt Output And Configuration Registers

## 1 Features

- Low standby current consumption of 1  $\mu$ A max
- I<sup>2</sup>C to Parallel port expander
- Open-drain active-low interrupt output
- Operating power-supply voltage range of 2.3 V to 5.5 V
- 5-V Tolerant I/O ports
- 400-kHz Fast I<sup>2</sup>C bus
- Three hardware address pins allow up to eight devices on the I<sup>2</sup>C/SMBus
- Allows up to 16 devices on the I<sup>2</sup>C/SMBus when used in conjunction with the [PCA9534](#)  
See [Section 5](#) for I<sup>2</sup>C Expander offerings
- Input and output configuration register
- Polarity inversion register
- Internal power-on reset
- Power-up with all channels configured as Inputs
- No glitch on power-up
- Noise filter on SCL and SDA inputs
- Latched outputs with high-current drive maximum capability for directly driving LEDs
- Latch-up performance exceeds 100 mA Per JESD 78, Class II
- ESD protection exceeds JESD 22
  - 2000-V Human-body Model (A114-A)
  - 200-V Machine Model (A115-A)
  - 1000-V Charged-device model (C101)

## 2 Description

This 8-bit I/O expander for the two-line bidirectional bus (I<sup>2</sup>C) is designed for 2.3-V to 5.5-V V<sub>CC</sub> operation. It provides general-purpose remote I/O expansion for most microcontroller families via the I<sup>2</sup>C interface [serial clock (SCL), serial data (SDA)].

The PCA9534A consists of one 8-bit configuration (input or output selection), input port, output port, and polarity inversion (active high or active low) register. At power on, the I/Os are configured as inputs. However, the system master can enable the I/Os as either inputs or outputs by writing to the I/O configuration bits. The data for each input or output is kept in the corresponding input or output register. The polarity of the input port register can be inverted with the polarity inversion register. All registers can be read by the system master.

The system master can reset the PCA9534A in the event of a timeout or other improper operation by utilizing the power-on reset feature, which puts the registers in their default state and initializes the I<sup>2</sup>C/SMBus state machine.

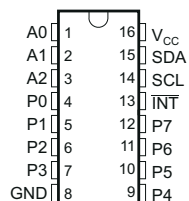
The PCA9534A open-drain interrupt ( $\overline{\text{INT}}$ ) output is activated when any input state differs from its corresponding input port register state and is used to indicate to the system master that an input state has changed.

### Device Information

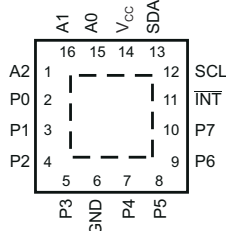
| PART NUMBER | PACKAGE <sup>(1)</sup> | BODY SIZE (NOM)   |
|-------------|------------------------|-------------------|
| PCA9534A    | SSOP (16)              | 6.20 mm × 5.30 mm |
|             | VQFN (16)              | 4.00 mm × 4.00 mm |
|             | QFN (16)               | 3.00 mm × 3.00 mm |

(1) For all available packages, see the orderable addendum at the end of the datasheet.

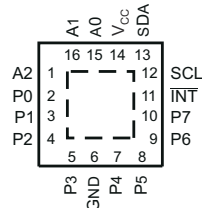
DB, DBQ, DGV, DW, OR PW PACKAGE  
(TOP VIEW)



RGV PACKAGE  
(TOP VIEW)



RGT PACKAGE  
(TOP VIEW)



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## 3 Revision History

| <b>Changes from Revision I (June 2014) to Revision J (March 2021)</b>                                                                                                          | <b>Page</b> |
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| • Moved the "Storage temperature range" to the <i>Absolute Maximum Ratings</i> .....                                                                                           | 6           |
| • Moved the "Package thermal impedance" to the <i>Thermal Resistance Characteristic</i> .....                                                                                  | 6           |
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| • Changed the V <sub>IL</sub> Low-level input voltage (A0, A1, A2, P7–P0) MAX value From: 0.8 V To: 0.3 x V <sub>CC</sub> in the <i>Recommended Operating Conditions</i> ..... | 6           |
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| • Changed the C <sub>i</sub> SCL Max value From: 5 pF To: 8 pF in the <i>Electrical Characteristics</i> .....                                                                  | 7           |
| • Changed the C <sub>io</sub> SDA Max value From: 6.5 pF To: 9.5 pF in the <i>Electrical Characteristics</i> .....                                                             | 7           |
| • Changed the t <sub>pv</sub> Output data valid MAX values From: 200 ns To 350 ns in the <i>Switchover Characteristics</i> .....                                               | 8           |
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## 4 Description (Continued)

The PCA9534A and PCA9534 are identical, except for their fixed I<sup>2</sup>C address. This allows for up to 16 of these devices (8 of each) on the same I<sup>2</sup>C bus.

$\overline{\text{INT}}$  can be connected to the interrupt input of a microcontroller. By sending an interrupt signal on this line, the remote I/O can inform the microcontroller if there is incoming data on its ports without having to communicate via the I<sup>2</sup>C bus. Thus, the PCA9534A can remain a simple slave device.

The device's outputs (latched) have high-current drive capability for directly driving LEDs. It has low current consumption.

Three hardware pins (A0, A1, and A2) are used to program and vary the fixed I<sup>2</sup>C address and allow up to eight devices to share the same I<sup>2</sup>C bus or SMBus.

The PCA9534A is pin-to-pin and I<sup>2</sup>C address compatible with the PCF8574A. However, software changes are required due to the enhancements in the PCA9534A over the PCF8574A.

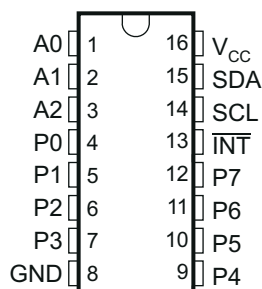
The PCA9534A is a low-power version of the PCA9554A. The only difference between the PCA9534A and PCA9554A is that the PCA9534A eliminates an internal I/O pullup resistor, which dramatically reduces power consumption in the standby mode when the I/Os are held low.

## 5 Device Comparison Table

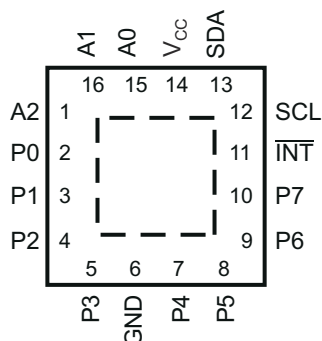
| DEVICE    | MAX FREQUENCY | I <sup>2</sup> C ADDRESS | V <sub>CC</sub> RANGE | NO. OF GPIOs | INTERRUPT OUTPUT | RESET INPUT | CONFIGURATION REGISTERS | 5-V TOLERANT | PUSH-PULL I/O TYPE | OPEN-DRAIN I/O TYPE | COMMENT                                                                                                                              |
|-----------|---------------|--------------------------|-----------------------|--------------|------------------|-------------|-------------------------|--------------|--------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| PCA6408   | 400           | 0100 00x                 | 1.65 to 5.5           | 8            | Yes              | Yes         | Yes                     | Yes          | Yes                | No                  | Power on reset, t <sub>f</sub> (fall time) > 100 ms and t <sub>r</sub> (ramp time) < 10 ms                                           |
| PCA6408   | 400           | 0100 00x                 | 1.65 to 5.5           | 8            | Yes              | Yes         | Yes                     | Yes          | Yes                | No                  | Unrestricted power on reset ramp/fall time. Both t <sub>f</sub> (fall time) and TRT (ramp time) can be between 0.1 ms and 2000 ms    |
| PCA6416   | 400           | 0100 00x                 | 1.65 to 5.5           | 16           | Yes              | Yes         | Yes                     | Yes          | Yes                | No                  | Power on reset, t <sub>f</sub> (fall time) > 100 ms and TRT (ramp time) < 10 ms                                                      |
| PCA6416 A | 400           | 0100 00x                 | 1.65 to 5.5           | 16           | Yes              | Yes         | Yes                     | Yes          | Yes                | No                  | Unrestricted power on reset ramp/fall time. Both t <sub>f</sub> (fall time) and TRT (ramp time) can be between 0.1 ms and 2000ms     |
| PCA6424   | 400           | 0100 00x                 | 1.65 to 5.5           | 24           | Yes              | Yes         | Yes                     | Yes          | Yes                | No                  | Power on reset, t <sub>f</sub> (fall time) > 100 ms and TRT (ramp time) < 10 ms                                                      |
| PCA9535   | 400           | 0100 xxx                 | 1.65 to 5.5           | 16           | Yes              | No          | Yes                     | Yes          | Yes                | No                  |                                                                                                                                      |
| PCA9539   | 400           | 1110 1xx                 | 1.65 to 5.5           | 16           | Yes              | Yes         | Yes                     | Yes          | Yes                | No                  |                                                                                                                                      |
| PCA9555   | 400           | 0100 xxx                 | 1.65 to 5.5           | 16           | Yes              | No          | Yes                     | Yes          | Yes                | No                  |                                                                                                                                      |
| PCA6107   | 400           | 0011 xxx                 | 2.3 to 5.5            | 8            | Yes              | Yes         | Yes                     | Yes          | Yes P1—P7 bits     | Yes P0 bit          | One open drain output; eight push pull outputs                                                                                       |
| PCA9534   | 400           | 0100 xxx                 | 2.3 to 5.5            | 8            | Yes              | No          | Yes                     | Yes          | Yes                | No                  | PCA9534 has a different slave address as the PCA9534A, allowing up to 16 devices '9534 type devices on the same I <sup>2</sup> C bus |
| PCA9534 A | 400           | 0111 xxx                 | 2.3 to 5.5            | 8            | Yes              | No          | Yes                     | Yes          | Yes                | No                  | PCA9534A has a different slave address as the PCA9534, allowing up to 16 devices '9534 type devices on the same I <sup>2</sup> C bus |
| PCA9535   | 400           | 0100 xxx                 | 2.3 to 5.5            | 16           | Yes              | No          | Yes                     | Yes          | Yes                | No                  |                                                                                                                                      |
| PCA9536   | 400           | 1000 001                 | 2.3 to 5.5            | 4            | No               | No          | Yes                     | Yes          | Yes                | No                  |                                                                                                                                      |
| PCA9538   | 400           | 1110 0xx                 | 2.3 to 5.5            | 8            | Yes              | Yes         | Yes                     | Yes          | Yes                | No                  |                                                                                                                                      |
| PCA9539   | 400           | 1110 1xx                 | 2.3 to 5.5            | 16           | Yes              | Yes         | Yes                     | Yes          | Yes                | No                  |                                                                                                                                      |
| PCA9554   | 400           | 0100 xxx                 | 2.3 to 5.5            | 8            | Yes              | No          | Yes                     | Yes          | Yes                | No                  |                                                                                                                                      |
| PCA9554 A | 400           | 0111 xxx                 | 2.3 to 5.5            | 8            | Yes              | No          | Yes                     | Yes          | Yes                | No                  |                                                                                                                                      |
| PCA9555   | 400           | 0100 xxx                 | 2.3 to 5.5            | 16           | Yes              | No          | Yes                     | Yes          | Yes                | No                  |                                                                                                                                      |
| PCA9557   | 400           | 0011 xxx                 | 2.3 to 5.5            | 8            | No               | Yes         | Yes                     | Yes          | Yes                | Yes                 |                                                                                                                                      |
| PCF8574   | 400           | 0100 xxx                 | 2.5 to 6.0            | 8            | Yes              | No          | No                      | Yes          | Yes                | No                  | PCA8574 has a different slave address as the PCA8574A, allowing up to 16 devices '9534 type devices on the same I <sup>2</sup> C bus |
| PCF8574 A | 400           | 0111 xxx                 | 2.5 to 6.0            | 8            | Yes              | No          | No                      | Yes          | Yes                | No                  | PCA8574A has a different slave address as the PCA8574, allowing up to 16 devices '9534 type devices on the same I <sup>2</sup> C bus |
| PCF8575   | 400           | 0100 xxx                 | 2.5 to 5.5            | 16           | Yes              | No          | No                      | Yes          | Yes                | No                  |                                                                                                                                      |
| PCF8575 C | 400           | 0100 xxx                 | 4.5 to 5.5            | 16           | Yes              | No          | No                      | Yes          | No                 | Yes                 |                                                                                                                                      |

## 6 Pin Configuration and Functions

DB, DBQ, DGV, DW, OR PW PACKAGE  
(TOP VIEW)



RGV PACKAGE  
(TOP VIEW)



RGT PACKAGE  
(TOP VIEW)

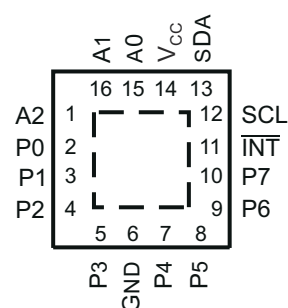


Table 6-1. Pin Functions

| NAME            | PIN                                                                       |                            | DESCRIPTION                                                             |
|-----------------|---------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------|
|                 | QSOP (DBQ),<br>SOIC (DW),<br>SSOP (DB),<br>TSSOP (PW), AND<br>TVSOP (DGV) | QFN (RGT) AND<br>QFN (RGV) |                                                                         |
| A0              | 1                                                                         | 15                         | Address input. Connect directly to V <sub>CC</sub> or ground.           |
| A1              | 2                                                                         | 16                         | Address input. Connect directly to V <sub>CC</sub> or ground.           |
| A2              | 3                                                                         | 1                          | Address input. Connect directly to V <sub>CC</sub> or ground.           |
| P0              | 4                                                                         | 2                          | P-port input/output. Push-pull design structure.                        |
| P1              | 5                                                                         | 3                          | P-port input/output. Push-pull design structure.                        |
| P2              | 6                                                                         | 4                          | P-port input/output. Push-pull design structure.                        |
| P3              | 7                                                                         | 5                          | P-port input/output. Push-pull design structure.                        |
| GND             | 8                                                                         | 6                          | Ground                                                                  |
| P4              | 9                                                                         | 7                          | P-port input/output. Push-pull design structure.                        |
| P5              | 10                                                                        | 8                          | P-port input/output. Push-pull design structure.                        |
| P6              | 11                                                                        | 9                          | P-port input/output. Push-pull design structure.                        |
| P7              | 12                                                                        | 10                         | P-port input/output. Push-pull design structure.                        |
| INT             | 13                                                                        | 11                         | Interrupt output. Connect to V <sub>CC</sub> through a pullup resistor. |
| SCL             | 14                                                                        | 12                         | Serial clock bus. Connect to V <sub>CC</sub> through a pullup resistor. |
| SDA             | 15                                                                        | 13                         | Serial data bus. Connect to V <sub>CC</sub> through a pullup resistor.  |
| V <sub>CC</sub> | 16                                                                        | 14                         | Supply voltage                                                          |

## 7 Specifications

### 7.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted) <sup>(1)</sup>

|                  |                                            | MIN                                                    | MAX  | UNIT |
|------------------|--------------------------------------------|--------------------------------------------------------|------|------|
| V <sub>CC</sub>  | Supply voltage range                       | −0.5                                                   | 6    | V    |
| V <sub>I</sub>   | Input voltage range <sup>(2)</sup>         | −0.5                                                   | 6    | V    |
| V <sub>O</sub>   | Output voltage range <sup>(2)</sup>        | −0.5                                                   | 6    | V    |
| I <sub>IK</sub>  | Input clamp current                        | V <sub>I</sub> < 0                                     | −20  | mA   |
| I <sub>OK</sub>  | Output clamp current                       | V <sub>O</sub> < 0                                     | −20  | mA   |
| I <sub>IOK</sub> | Input/output clamp current                 | V <sub>O</sub> < 0 or V <sub>O</sub> > V <sub>CC</sub> | ±20  | mA   |
| I <sub>OL</sub>  | Continuous output low current              | V <sub>O</sub> = 0 to V <sub>CC</sub>                  | 50   | mA   |
| I <sub>OH</sub>  | Continuous output high current             | V <sub>O</sub> = 0 to V <sub>CC</sub>                  | −50  | mA   |
| I <sub>CC</sub>  | Continuous current through GND             |                                                        | −250 | mA   |
|                  | Continuous current through V <sub>CC</sub> |                                                        | 160  |      |
| T <sub>stg</sub> | Storage temperature range                  | −65                                                    | 150  | °C   |

(1) Stresses beyond those listed under "absolute maximum ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under "recommended operating conditions" is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

(2) The input negative-voltage and output voltage ratings may be exceeded if the input and output current ratings are observed.

### 7.2 ESD Ratings

|                    |                         |                                                                                          | MIN | MAX  | UNIT |
|--------------------|-------------------------|------------------------------------------------------------------------------------------|-----|------|------|
| V <sub>(ESD)</sub> | Electrostatic discharge | Human body model (HBM), per ANSI/ESDA/JEDEC JS-001, all pins <sup>(1)</sup>              | 0   | 2000 | V    |
|                    |                         | Charged device model (CDM), per JEDEC specification JESD22-C101, all pins <sup>(2)</sup> | 0   | 1000 |      |

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

### 7.3 Recommended Operating Conditions

|                 |                                |              | MIN                   | MAX                   | UNIT |
|-----------------|--------------------------------|--------------|-----------------------|-----------------------|------|
| V <sub>CC</sub> | Supply voltage                 |              | 2.3                   | 5.5                   | V    |
| V <sub>IH</sub> | High-level input voltage       | SCL, SDA     | 0.7 × V <sub>CC</sub> | V <sub>CC</sub>       | V    |
|                 |                                | A2–A0, P7–P0 | 0.7 × V <sub>CC</sub> | 5.5                   |      |
| V <sub>IL</sub> | Low-level input voltage        | SCL, SDA     | −0.5                  | 0.3 × V <sub>CC</sub> | V    |
|                 |                                | A2–A0, P7–P0 | −0.5                  | 0.3 × V <sub>CC</sub> |      |
| I <sub>OH</sub> | High-level output current      | P7–P0        |                       | −10                   | mA   |
| I <sub>OL</sub> | Low-level output current       | P7–P0        |                       | 25                    | mA   |
| T <sub>A</sub>  | Operating free-air temperature |              | −40                   | 85                    | °C   |

### 7.4 Thermal Resistance Characteristics

| THERMAL METRIC <sup>(1)</sup> |                                        | PCA9535      |               |                |              |               |               |               | UNIT |
|-------------------------------|----------------------------------------|--------------|---------------|----------------|--------------|---------------|---------------|---------------|------|
|                               |                                        | DB<br>(SSOP) | DBQ<br>(SSOP) | DVG<br>(TVSOP) | DW<br>(SOIC) | PW<br>(TSSOP) | RGT<br>(VQFN) | RVE<br>(VQFN) |      |
|                               |                                        | 16 Pins      | 16 Pins       | 16 Pins        | 16 Pins      | 16 Pins       | 16 Pins       | 16 Pins       |      |
| R <sub>θJA</sub>              | Junction-to-ambient thermal resistance | 82           | 90            | 86             | 92.2         | 122           | 63.2          | 51            | °C/W |

(1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC package thermal metrics](#) application report.

## 7.5 Electrical Characteristics

over operating free-air temperature range (unless otherwise noted)

| PARAMETER         |                                                 | TEST CONDITIONS                                                                                       | V <sub>CC</sub> | MIN  | TYP <sup>(1)</sup> | MAX | UNIT |
|-------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------|------|--------------------|-----|------|
| V <sub>IK</sub>   | Input diode clamp voltage                       | I <sub>I</sub> = –18 mA                                                                               | 2.3 V to 5.5 V  | –1.2 |                    |     | V    |
| V <sub>PORR</sub> | Power-on reset voltage, V <sub>CC</sub> rising  | V <sub>I</sub> = V <sub>CC</sub> or GND, I <sub>O</sub> = 0                                           |                 |      | 1.2                | 1.5 | V    |
| V <sub>PORF</sub> | Power-on reset voltage, V <sub>CC</sub> falling | V <sub>I</sub> = V <sub>CC</sub> or GND, I <sub>O</sub> = 0                                           |                 | 0.75 | 1                  |     |      |
| V <sub>OH</sub>   | P-port high-level output voltage <sup>(2)</sup> | I <sub>OH</sub> = –8 mA                                                                               | 2.3 V           | 1.8  |                    |     | V    |
|                   |                                                 |                                                                                                       | 3 V             | 2.6  |                    |     |      |
|                   |                                                 |                                                                                                       | 4.5 V           | 4.1  |                    |     |      |
|                   |                                                 |                                                                                                       | 4.75 V          | 4.1  |                    |     |      |
|                   |                                                 | I <sub>OH</sub> = –10 mA                                                                              | 2.3 V           | 1.7  |                    |     |      |
|                   |                                                 |                                                                                                       | 3 V             | 2.5  |                    |     |      |
|                   |                                                 |                                                                                                       | 4.5 V           | 4    |                    |     |      |
|                   |                                                 |                                                                                                       | 4.75 V          | 4    |                    |     |      |
| I <sub>OL</sub>   | SDA                                             | V <sub>OL</sub> = 0.4 V                                                                               | 2.3 V to 5.5 V  | 3    | 8                  |     | mA   |
|                   | P port <sup>(3)</sup>                           | V <sub>OL</sub> = 0.5 V                                                                               | 2.3 V           | 8    | 10                 |     |      |
|                   |                                                 |                                                                                                       | 3 V             | 8    | 14                 |     |      |
|                   |                                                 |                                                                                                       | 4.5 V           | 8    | 17                 |     |      |
|                   |                                                 |                                                                                                       | 4.75 V          | 8    | 35                 |     |      |
|                   |                                                 | V <sub>OL</sub> = 0.7 V                                                                               | 2.3 V           | 10   | 13                 |     |      |
|                   |                                                 |                                                                                                       | 3 V             | 10   | 19                 |     |      |
|                   |                                                 |                                                                                                       | 4.5 V           | 10   | 24                 |     |      |
|                   |                                                 |                                                                                                       | 4.75 V          | 10   | 45                 |     |      |
|                   | INT                                             | V <sub>OL</sub> = 0.4 V                                                                               | 2.3 V to 5.5 V  | 3    | 10                 |     |      |
| I <sub>I</sub>    | SCL, SDA                                        | V <sub>I</sub> = V <sub>CC</sub> or GND                                                               | 2.3 V to 5.5 V  |      |                    | ±1  | μA   |
|                   | A2–A0                                           |                                                                                                       |                 |      |                    | ±1  |      |
| I <sub>IH</sub>   | P port                                          | V <sub>I</sub> = V <sub>CC</sub>                                                                      | 2.3 V to 5.5 V  |      |                    | 1   | μA   |
| I <sub>IL</sub>   | P port                                          | V <sub>I</sub> = GND                                                                                  | 2.3 V to 5.5 V  |      |                    | –1  | μA   |
| I <sub>CC</sub>   | Operating mode                                  | V <sub>I</sub> = V <sub>CC</sub> or GND, I <sub>O</sub> = 0, I/O = inputs, f <sub>scl</sub> = 400 kHz | 5.5 V           |      | 104                | 175 | μA   |
|                   |                                                 |                                                                                                       | 3.6 V           |      | 50                 | 90  |      |
|                   |                                                 |                                                                                                       | 2.7 V           |      | 20                 | 65  |      |
|                   |                                                 | V <sub>I</sub> = V <sub>CC</sub> or GND, I <sub>O</sub> = 0, I/O = inputs, f <sub>scl</sub> = 100 kHz | 5.5 V           |      | 60                 | 150 |      |
|                   |                                                 |                                                                                                       | 3.6 V           |      | 15                 | 40  |      |
|                   |                                                 |                                                                                                       | 2.7 V           |      | 8                  | 20  |      |
|                   | Standby mode                                    | V <sub>I</sub> = GND, I <sub>O</sub> = 0, I/O = inputs, f <sub>scl</sub> = 0 kHz                      | 5.5 V           |      | 1.5                | 8.7 |      |
|                   |                                                 |                                                                                                       | 3.6 V           |      | 0.9                | 4   |      |
|                   |                                                 |                                                                                                       | 2.7 V           |      | 0.6                | 3   |      |
| ΔI <sub>CC</sub>  | Additional current in standby mode              | One input at V <sub>CC</sub> – 0.6 V, Other inputs at V <sub>CC</sub> or GND                          | 2.3 V to 5.5 V  |      |                    | 1.5 | mA   |
|                   |                                                 | All LED I/Os at V <sub>I</sub> = 4.3 V, f <sub>scl</sub> = 0 kHz                                      | 5.5 V           |      |                    | 1   |      |
| C <sub>I</sub>    | SCL                                             | V <sub>I</sub> = V <sub>CC</sub> or GND                                                               | 2.3 V to 5.5 V  |      | 4                  | 8   | pF   |
| C <sub>io</sub>   | SDA                                             | V <sub>IO</sub> = V <sub>CC</sub> or GND                                                              | 2.3 V to 5.5 V  |      | 5.5                | 9.5 | pF   |
|                   | P port                                          |                                                                                                       |                 |      | 8                  | 9.5 |      |

(1) All typical values are at nominal supply voltage (2.5-V, 3.3-V, or 5-V V<sub>CC</sub>) and T<sub>A</sub> = 25°C.

(2) The total current sourced by all I/Os must be limited to 85 mA.

(3) Each I/O must be externally limited to a maximum of 25 mA, and the P port (P7–P0) must be limited to a maximum current of 200 mA.

## 7.6 I<sup>2</sup>C Interface Timing Requirements

over operating free-air temperature range (unless otherwise noted) (see [Figure 8-1](#))

|                       |                                                          |                                          | STANDARD MODE<br>I <sup>2</sup> C BUS |      | FAST MODE<br>I <sup>2</sup> C BUS     |     | UNIT |
|-----------------------|----------------------------------------------------------|------------------------------------------|---------------------------------------|------|---------------------------------------|-----|------|
|                       |                                                          |                                          | MIN                                   | MAX  | MIN                                   | MAX |      |
| f <sub>scl</sub>      | I <sup>2</sup> C clock frequency                         |                                          | 0                                     | 100  | 0                                     | 400 | kHz  |
| t <sub>sch</sub>      | I <sup>2</sup> C clock high time                         |                                          | 4                                     |      | 0.6                                   |     | μs   |
| t <sub>scl</sub>      | I <sup>2</sup> C clock low time                          |                                          | 4.7                                   |      | 1.3                                   |     | μs   |
| t <sub>sp</sub>       | I <sup>2</sup> C spike time                              |                                          |                                       | 50   |                                       | 50  | ns   |
| t <sub>sds</sub>      | I <sup>2</sup> C serial-data setup time                  |                                          | 250                                   |      | 100                                   |     | ns   |
| t <sub>sdh</sub>      | I <sup>2</sup> C serial-data hold time                   |                                          | 0                                     |      | 0                                     |     | ns   |
| t <sub>icr</sub>      | I <sup>2</sup> C input rise time                         |                                          |                                       | 1000 | 20 + 0.1C <sub>b</sub> <sup>(1)</sup> | 300 | ns   |
| t <sub>icf</sub>      | I <sup>2</sup> C input fall time                         |                                          |                                       | 300  | 20 + 0.1C <sub>b</sub> <sup>(1)</sup> | 300 | ns   |
| t <sub>ocf</sub>      | I <sup>2</sup> C output fall time                        | 10-pF to 400-pF bus                      |                                       | 300  | 20 + 0.1C <sub>b</sub> <sup>(1)</sup> | 300 | ns   |
| t <sub>buf</sub>      | I <sup>2</sup> C bus free time between stop and start    |                                          | 4.7                                   |      | 1.3                                   |     | μs   |
| t <sub>sts</sub>      | I <sup>2</sup> C start or repeated start condition setup |                                          | 4.7                                   |      | 0.6                                   |     | μs   |
| t <sub>sth</sub>      | I <sup>2</sup> C start or repeated start condition hold  |                                          | 4                                     |      | 0.6                                   |     | μs   |
| t <sub>sps</sub>      | I <sup>2</sup> C stop condition setup                    |                                          | 4                                     |      | 0.6                                   |     | μs   |
| t <sub>vd(data)</sub> | Valid data time                                          | SCL low to SDA output valid              | 300                                   |      | 50                                    |     | ns   |
| t <sub>vd(ack)</sub>  | Valid data time of ACK condition                         | ACK signal from SCL low to SDA (out) low | 0.3                                   | 3.45 | 0.1                                   | 0.9 | μs   |
| C <sub>b</sub>        | I <sup>2</sup> C bus capacitive load                     |                                          |                                       | 400  |                                       | 400 | ns   |

(1) C<sub>b</sub> = total capacitive of one bus in pF

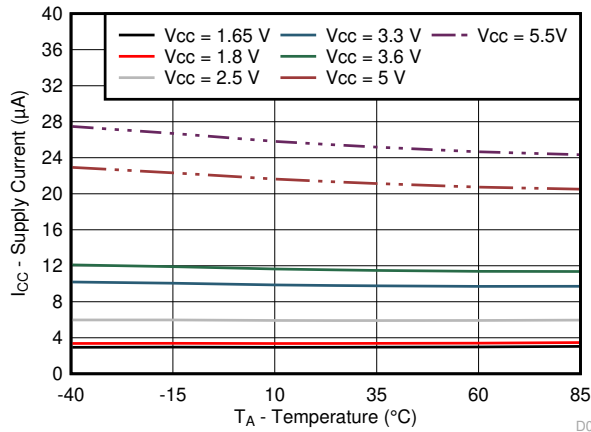
## 7.7 Switching Characteristics

over operating free-air temperature range (unless otherwise noted) (see [Figure 8-2](#) and [Figure 8-3](#))

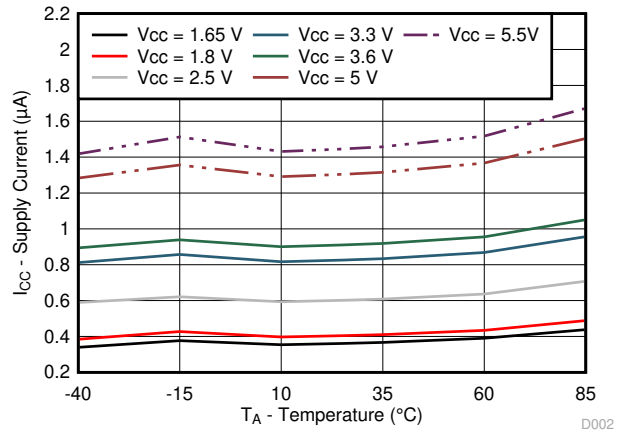
| PARAMETER       |                            | FROM<br>(INPUT) | TO<br>(OUTPUT) | STANDARD MODE<br>I <sup>2</sup> C BUS |     | FAST MODE<br>I <sup>2</sup> C BUS |     | UNIT |
|-----------------|----------------------------|-----------------|----------------|---------------------------------------|-----|-----------------------------------|-----|------|
|                 |                            |                 |                | MIN                                   | MAX | MIN                               | MAX |      |
| t <sub>iv</sub> | Interrupt valid time       | P port          | INT            |                                       | 4   |                                   | 4   | μs   |
| t <sub>ir</sub> | Interrupt reset delay time | SCL             | INT            |                                       | 4   |                                   | 4   | μs   |
| t <sub>pV</sub> | Output data valid          | SCL             | P7–P0          |                                       | 350 |                                   | 350 | ns   |
| t <sub>ps</sub> | Input data setup time      | P port          | SCL            | 100                                   |     | 100                               |     | ns   |
| t <sub>ph</sub> | Input data hold time       | P port          | SCL            | 1                                     |     | 1                                 |     | μs   |

## 7.8 Typical Characteristics

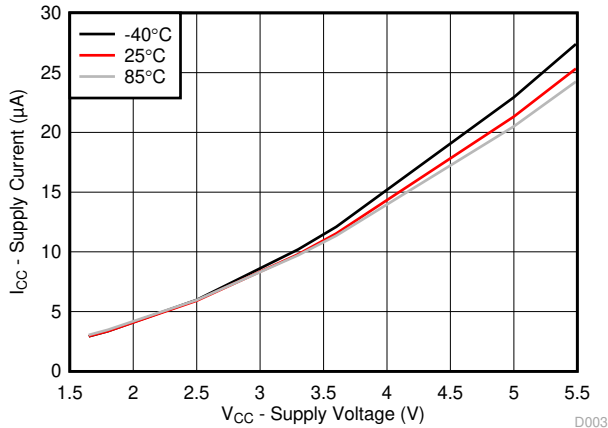
$T_A = 25^\circ\text{C}$  (unless otherwise noted)



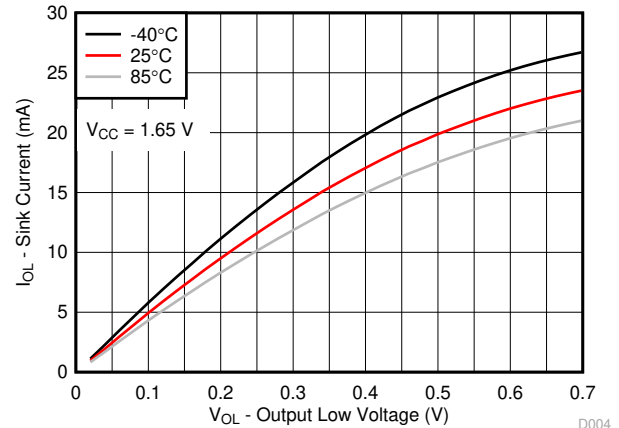
**Figure 7-1. Supply Current vs Temperature for Different Supply Voltage ( $V_{CC}$ )**



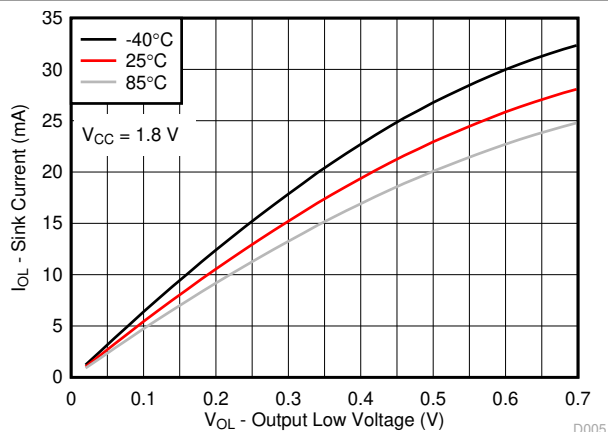
**Figure 7-2. Standby Supply Current vs Temperature for Different Supply Voltage ( $V_{CC}$ )**



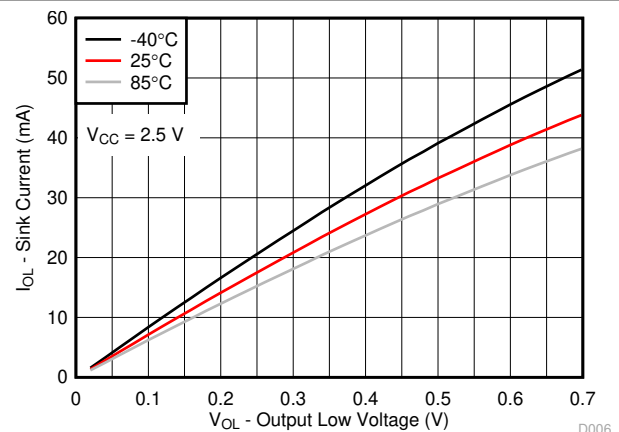
**Figure 7-3. Supply Current vs Supply Voltage for Different Temperature ( $T_A$ )**



**Figure 7-4. I/O Sink Current vs Output Low Voltage for Different Temperature ( $T_A$ ) for  $V_{CC} = 1.65\text{ V}$**



**Figure 7-5. I/O Sink Current vs Output Low Voltage for Different Temperature ( $T_A$ ) for  $V_{CC} = 1.8\text{ V}$**



**Figure 7-6. I/O Sink Current vs Output Low Voltage for Different Temperature ( $T_A$ ) for  $V_{CC} = 2.5\text{ V}$**

## 7.8 Typical Characteristics (continued)

$T_A = 25^\circ\text{C}$  (unless otherwise noted)

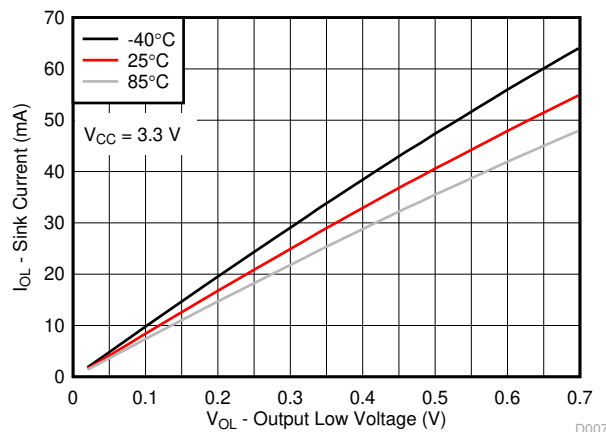


Figure 7-7. I/O Sink Current vs Output Low Voltage for Different Temperature ( $T_A$ ) for  $V_{CC} = 3.3\text{ V}$

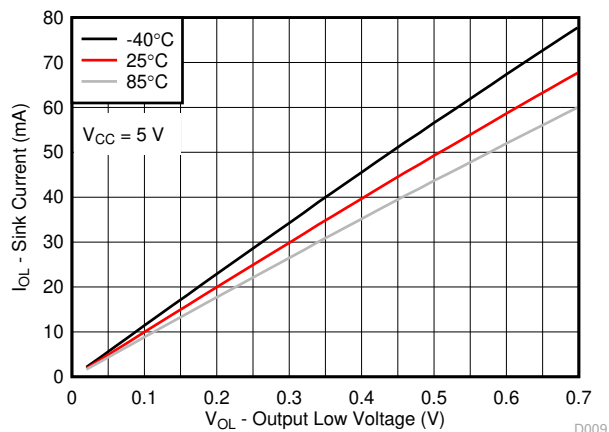


Figure 7-8. I/O Sink Current vs Output Low Voltage for Different Temperature ( $T_A$ ) for  $V_{CC} = 5\text{ V}$

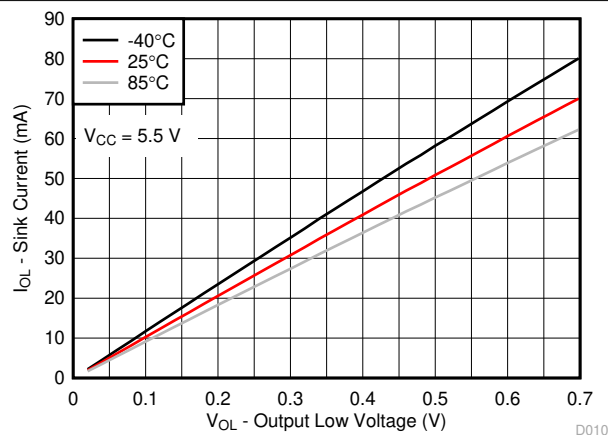


Figure 7-9. I/O Sink Current vs Output Low Voltage for Different Temperature ( $T_A$ ) for  $V_{CC} = 5.5\text{ V}$

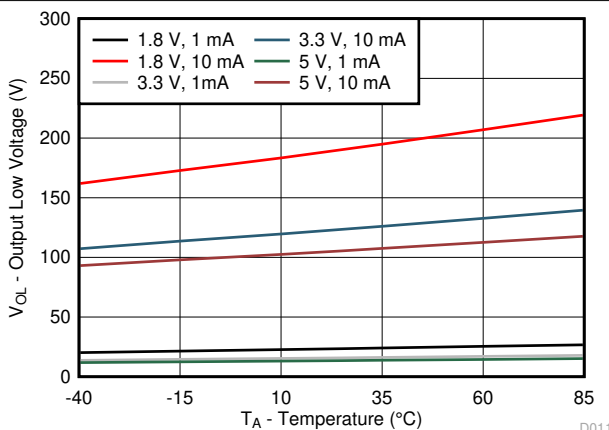


Figure 7-10. I/O Low Voltage vs Temperature for Different  $V_{CC}$  and  $I_{OL}$

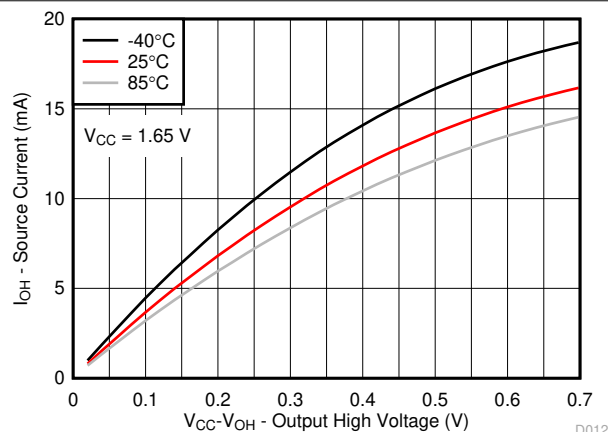


Figure 7-11. I/O Source Current vs Output High Voltage for Different Temperature ( $T_A$ ) for  $V_{CC} = 1.65\text{ V}$

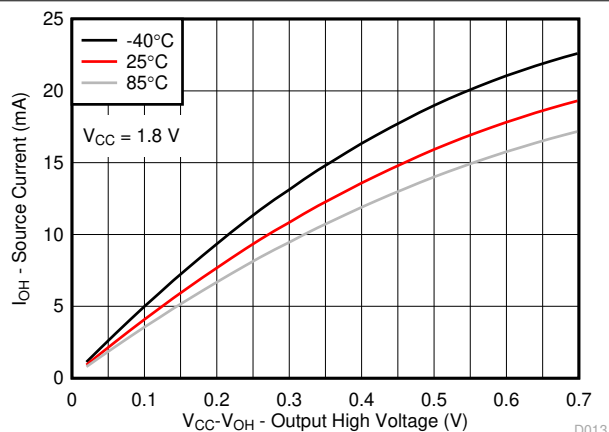
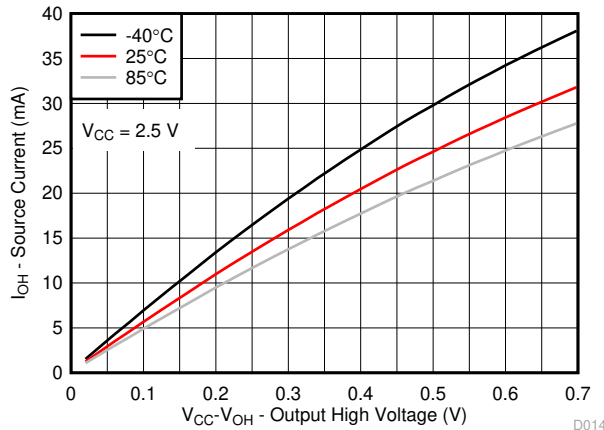


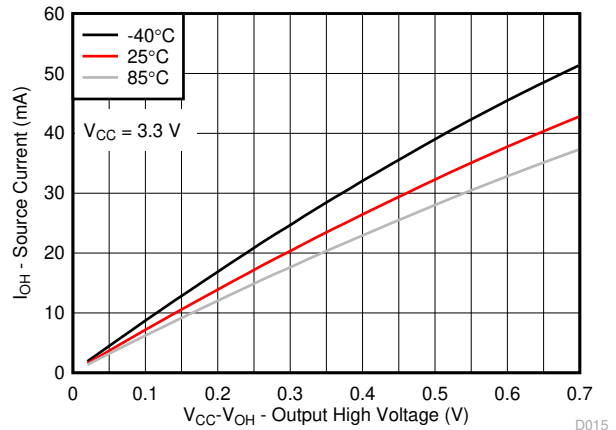
Figure 7-12. I/O Source Current vs Output High Voltage for Different Temperature ( $T_A$ ) for  $V_{CC} = 1.8\text{ V}$

## 7.8 Typical Characteristics (continued)

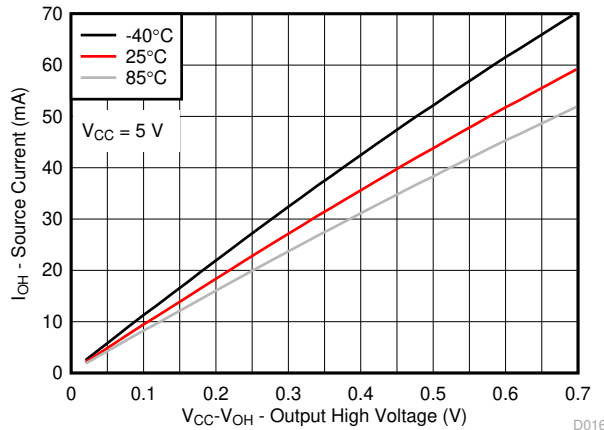
$T_A = 25^\circ\text{C}$  (unless otherwise noted)



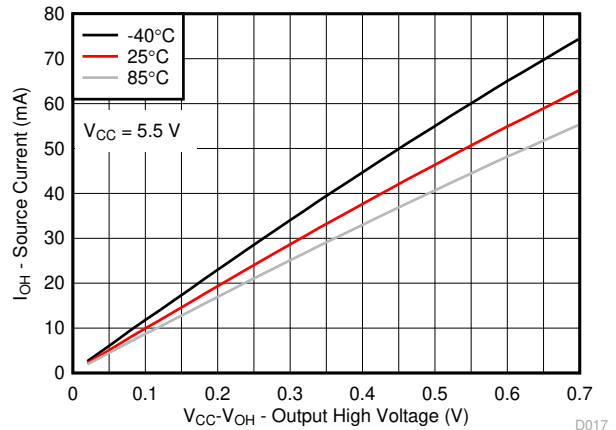
**Figure 7-13. I/O Source Current vs Output High Voltage for Different Temperature ( $T_A$ ) for  $V_{CC} = 2.5\text{ V}$**



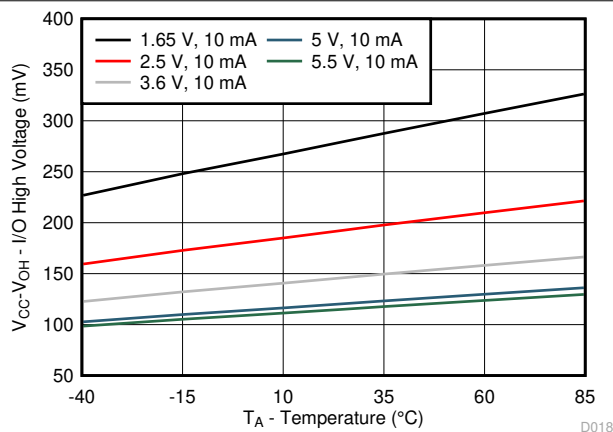
**Figure 7-14. I/O Source Current vs Output High Voltage for Different Temperature ( $T_A$ ) for  $V_{CC} = 3.3\text{ V}$**



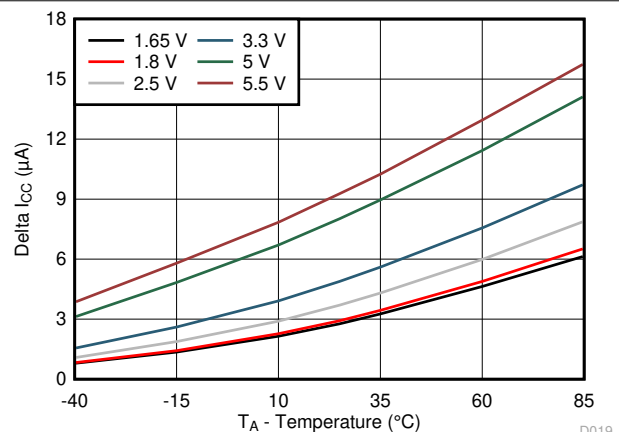
**Figure 7-15. I/O Source Current vs Output High Voltage for Different Temperature ( $T_A$ ) for  $V_{CC} = 5\text{ V}$**



**Figure 7-16. I/O Source Current vs Output High Voltage for Different Temperature ( $T_A$ ) for  $V_{CC} = 5.5\text{ V}$**

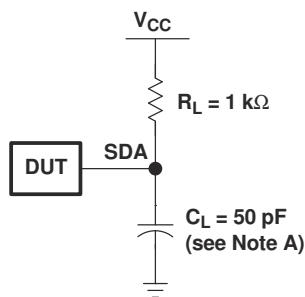


**Figure 7-17.  $V_{CC} - V_{OH}$  Voltage vs Temperature for Different  $V_{CC}$**

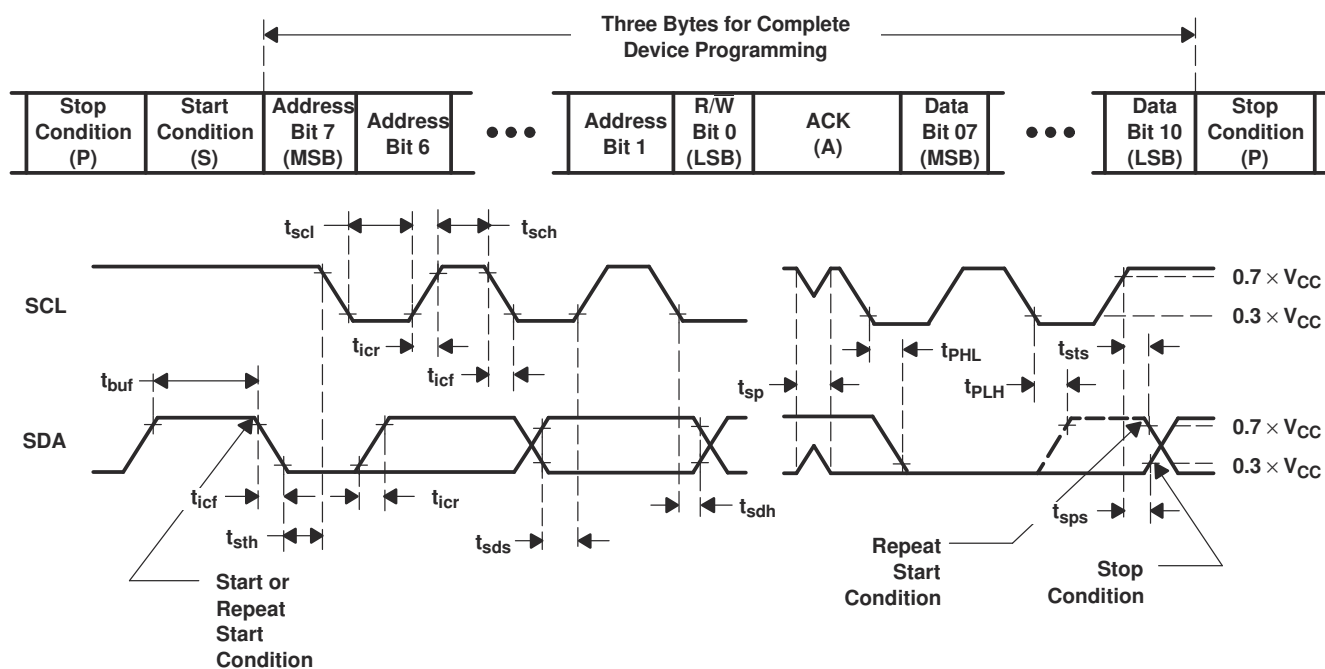


**Figure 7-18.  $\Delta I_{CC}$  vs Temperature for Different  $V_{CC}$  ( $V_I = V_{CC} - 0.6\text{ V}$ )**

## 8 Parameter Measurement Information



SDA LOAD CONFIGURATION

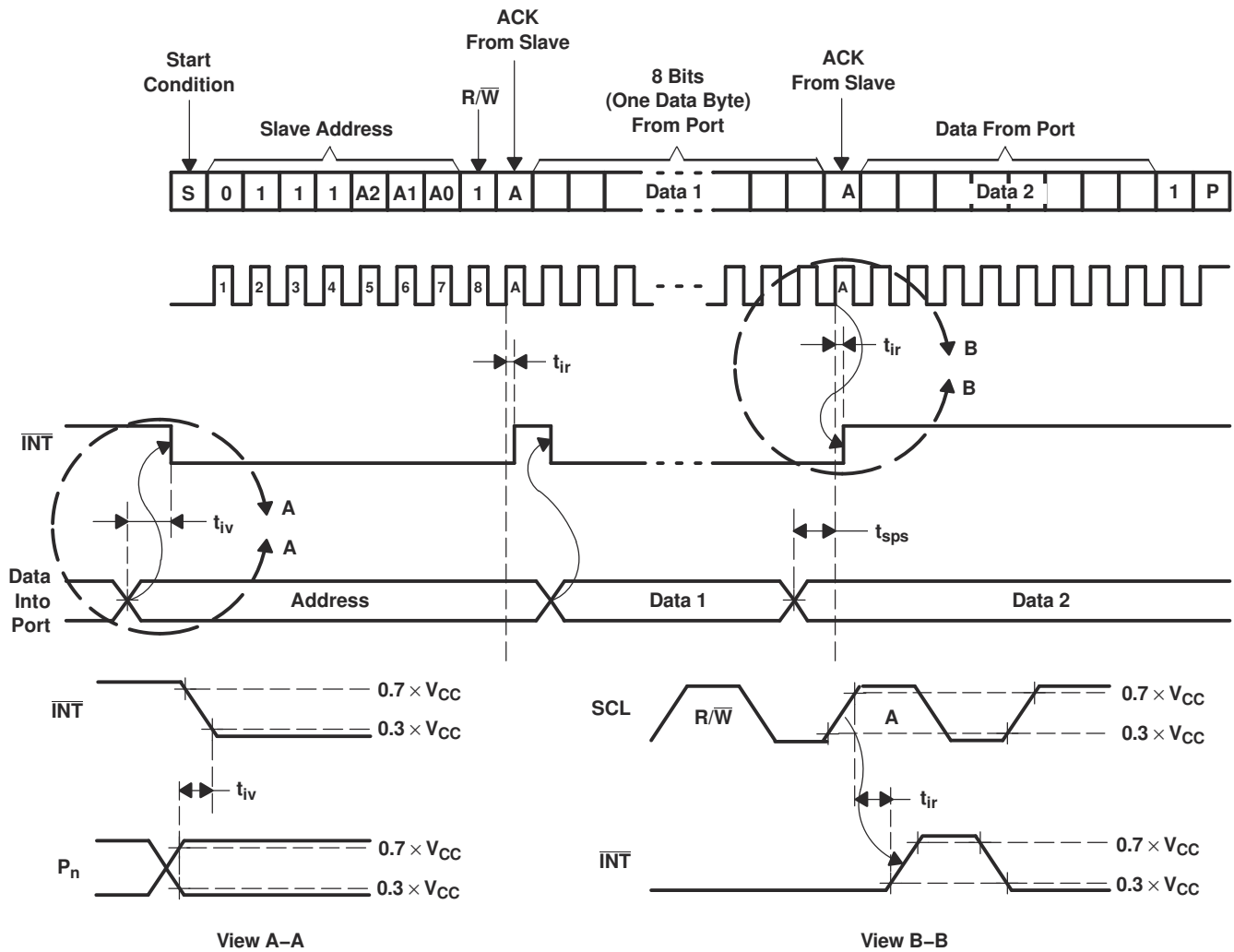
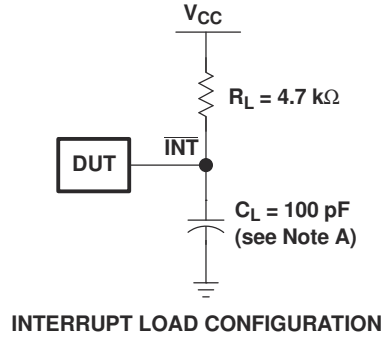


VOLTAGE WAVEFORMS

| BYTE | DESCRIPTION              |
|------|--------------------------|
| 1    | I <sup>2</sup> C address |
| 2, 3 | P-port data              |

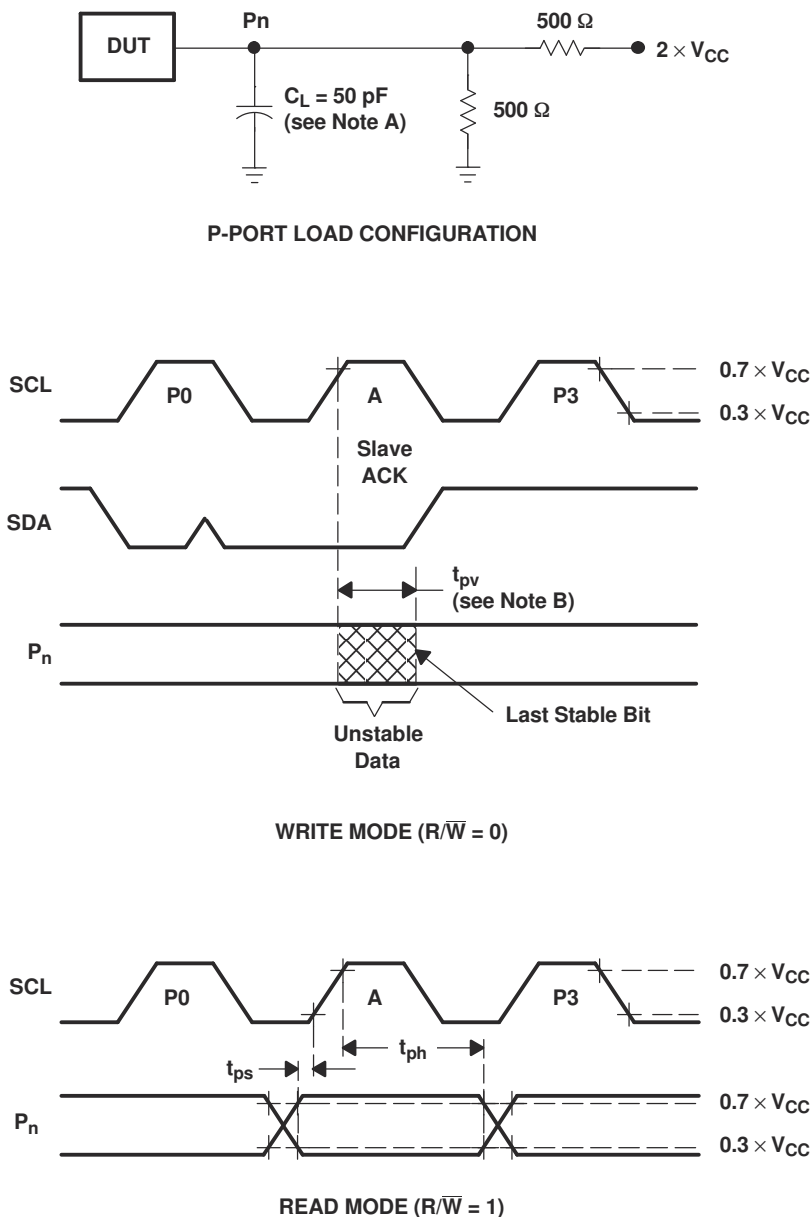
- A.  $C_L$  includes probe and jig capacitance.  
 B. All inputs are supplied by generators having the following characteristics: PRR  $\leq 10\text{ MHz}$ ,  $Z_O = 50\ \Omega$ ,  $t_r/t_f \leq 30\text{ ns}$ .  
 C. All parameters and waveforms are not applicable to all devices.

**Figure 8-1. I<sup>2</sup>C Interface Load Circuit And Voltage Waveforms**



- A.  $C_L$  includes probe and jig capacitance.  
 B. All inputs are supplied by generators having the following characteristics:  $PRR \leq 10 \text{ MHz}$ ,  $Z_O = 50 \Omega$ ,  $t_r/t_f \leq 30 \text{ ns}$ .  
 C. All parameters and waveforms are not applicable to all devices.

**Figure 8-2. Interrupt Load Circuit And Voltage Waveforms**

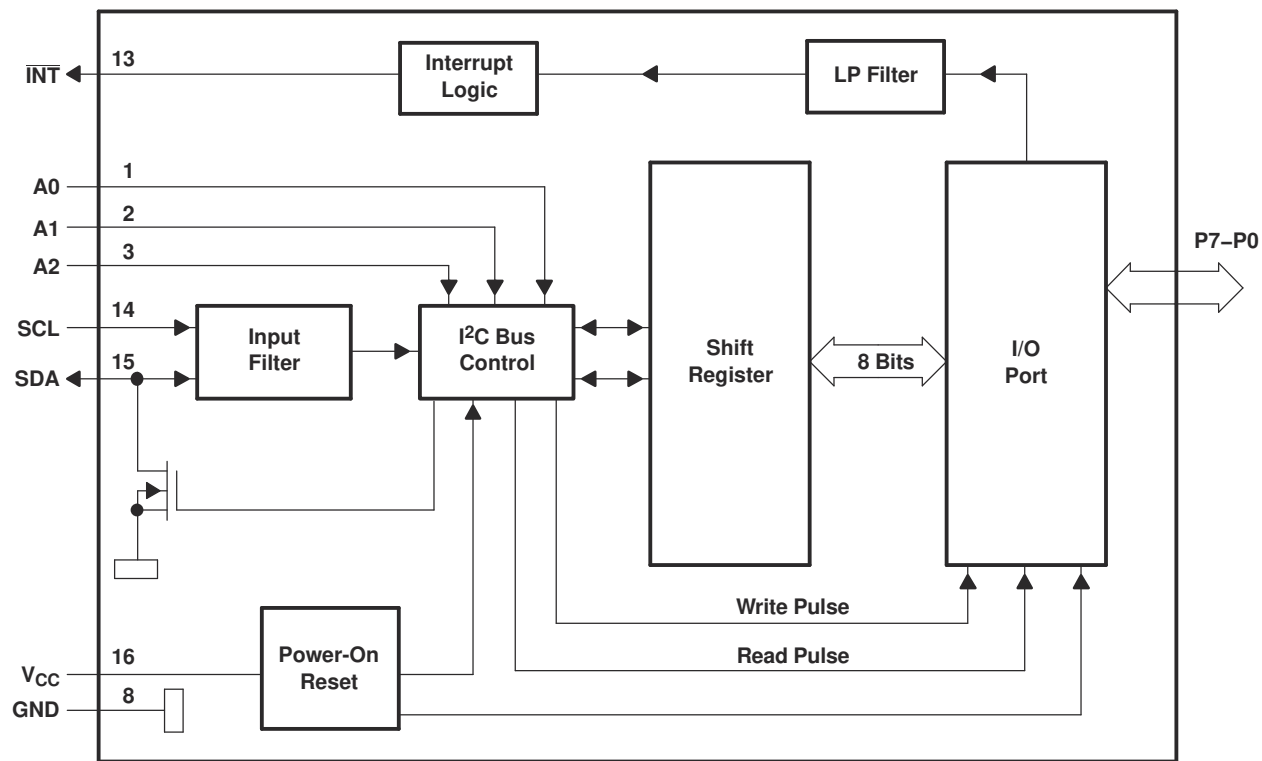


- A. C<sub>L</sub> includes probe and jig capacitance.
- B. t<sub>pv</sub> is measured from 0.7 × V<sub>CC</sub> on SCL to 50% I/O (P<sub>n</sub>) output.
- C. All inputs are supplied by generators having the following characteristics: PRR ≤ 10 MHz, Z<sub>O</sub> = 50 Ω, t<sub>r</sub>/t<sub>f</sub> ≤ 30 ns.
- D. The outputs are measured one at a time, with one transition per measurement.
- E. All parameters and waveforms are not applicable to all devices.

**Figure 8-3. P-Port Load Circuit And Voltage Waveforms**

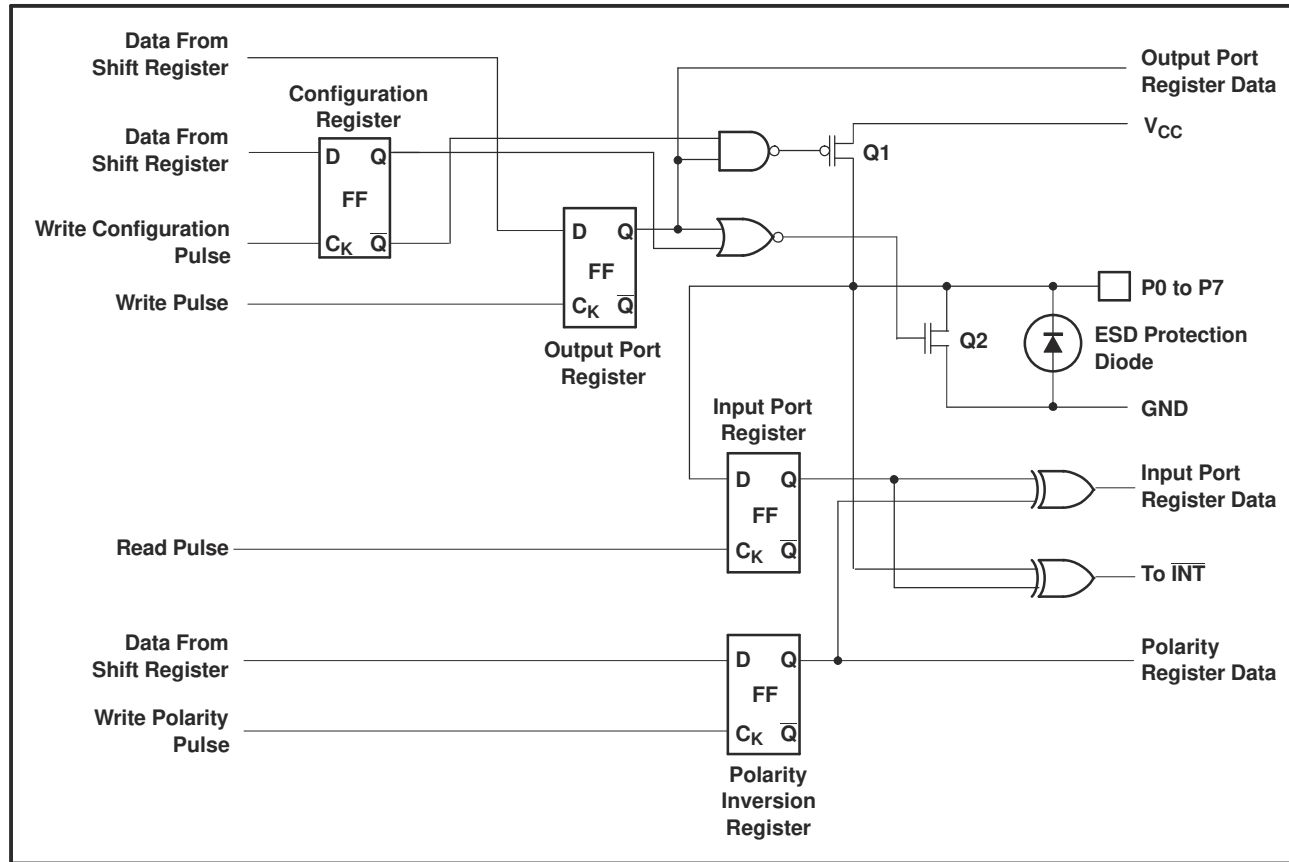
## 9 Detailed Description

### 9.1 Functional Block Diagram



- A. Pin numbers shown are for DB, DBQ, DGV, DW, or PW package.  
B. All I/Os are set to inputs at reset.

**Figure 9-1. Logic Diagram (Positive Logic)**



A. At power-on reset, all registers return to default values.

**Figure 9-2. Simplified Schematic Of P0 To P7**

## 9.2 Device Functional Modes

### 9.2.1 Power-On Reset

When power (from 0 V) is applied to  $V_{CC}$ , an internal power-on reset holds the PCA9534A in a reset condition until  $V_{CC}$  has reached  $V_{POR}$ . At that point, the reset condition is released and the PCA9534A registers and I<sup>2</sup>C/SMBus state machine initialize to their default states. After that,  $V_{CC}$  must be lowered to below 0.2 V and then back up to the operating voltage for a power-reset cycle.

### 9.2.2 I/O Port

When an I/O is configured as an input, FETs Q1 and Q2 (in [Figure 9-2](#)) are off, creating a high-impedance input. The input voltage may be raised above  $V_{CC}$  to a maximum of 5.5 V.

If the I/O is configured as an output, Q1 or Q2 is enabled, depending on the state of the output port register. In this case, there are low-impedance paths between the I/O pin and either  $V_{CC}$  or GND. The external voltage applied to this I/O pin should not exceed the recommended levels for proper operation.

### 9.2.3 Interrupt Output ( $\overline{INT}$ )

An interrupt is generated by any rising or falling edge of the port inputs in the input mode. After time,  $t_{IV}$ , the signal  $\overline{INT}$  is valid. Resetting the interrupt circuit is achieved when data on the port is changed to the original setting, data is read from the port that generated the interrupt. Resetting occurs in the read mode at the acknowledge (ACK) or not acknowledge (NACK) bit after the rising edge of the SCL signal.

Interrupts that occur during the ACK or NACK clock pulse can be lost (or be very short) due to the resetting of the interrupt during this pulse. Each change of the I/Os after resetting is detected and is transmitted as  $\overline{INT}$ . Writing to another device does not affect the interrupt circuit, and a pin configured as an output cannot cause

an interrupt. Changing an I/O from an output to an input may cause a false interrupt to occur, if the state of the pin does not match the contents of the Input Port register. Because each 8-pin port is read independently, the interrupt caused by port 0 is not cleared by a read of port 1 or vice versa.

The  $\overline{\text{INT}}$  output has an open-drain structure and requires pull-up resistor to  $V_{CC}$ .

### 9.2.3.1 Interrupt Errata

#### 9.2.3.1.1 Description

The INT will be improperly de-asserted if the following two conditions occur:

1. The last I<sup>2</sup>C command byte (register pointer) written to the device was 00h.

---

#### Note

This generally means the last operation with the device was a Read of the input register. However, the command byte may have been written with 00h without ever going on to read the input register. After reading from the device, if no other command byte written, it will remain 00h.

2. Any other slave device on the I<sup>2</sup>C bus acknowledges an address byte with the R/W bit set high

#### 9.2.3.1.2 System Impact

Can cause improper interrupt handling as the Master will see the interrupt as being cleared.

#### 9.2.3.1.3 System Workaround

Minor software change: User must change command byte to something besides 00h after a Read operation to the PCA9534A device or before reading from another slave device.

---

#### Note

Software change will be compatible with other versions (competition and TI redesigns) of this device.

## 9.3 Programming

### 9.3.1 I<sup>2</sup>C Interface

The bidirectional I<sup>2</sup>C bus consists of the serial clock (SCL) and serial data (SDA) lines. Both lines must be connected to a positive supply through a pull-up resistor when connected to the output stages of a device. Data transfer may be initiated only when the bus is not busy.

I<sup>2</sup>C communication with this device is initiated by a master sending a start condition, a high-to-low transition on the SDA input/output while the SCL input is high (see [Figure 9-3](#)). After the start condition, the device address byte is sent, MSB first, including the data direction bit (R/W).

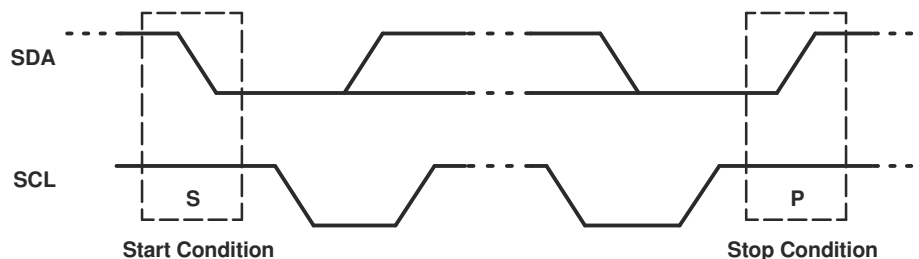
After receiving the valid address byte, this device responds with an acknowledge (ACK), a low on the SDA input/output during the high of the ACK-related clock pulse. The address inputs (A0–A2) of the slave device must not be changed between the start and the stop conditions.

On the I<sup>2</sup>C bus, only one data bit is transferred during each clock pulse. The data on the SDA line must remain stable during the high pulse of the clock period, as changes in the data line at this time are interpreted as control commands (start or stop) (see [Figure 9-4](#)).

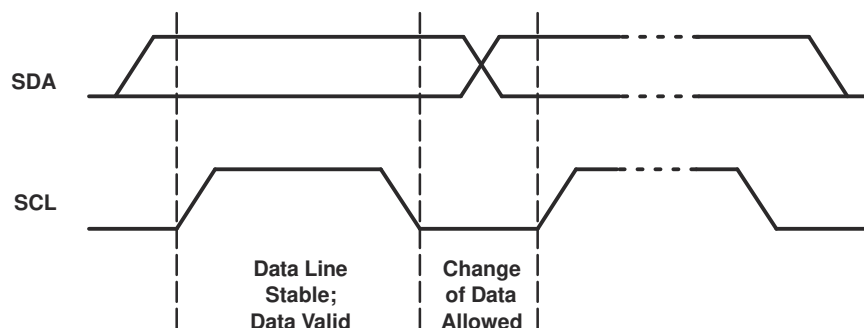
A stop condition, a low-to-high transition on the SDA input/output while the SCL input is high, is sent by the master (see [Figure 9-3](#)).

Any number of data bytes can be transferred from the transmitter to receiver between the start and the stop conditions. Each byte of eight bits is followed by one ACK bit. The transmitter must release the SDA line before the receiver can send an ACK bit. The device that acknowledges must pull down the SDA line during the ACK clock pulse so that the SDA line is stable low during the high pulse of the ACK-related clock period (see [Figure 9-5](#)). When a slave receiver is addressed, it must generate an ACK after each byte is received. Similarly, the master must generate an ACK after each byte that it receives from the slave transmitter. Setup and hold times must be met to ensure proper operation.

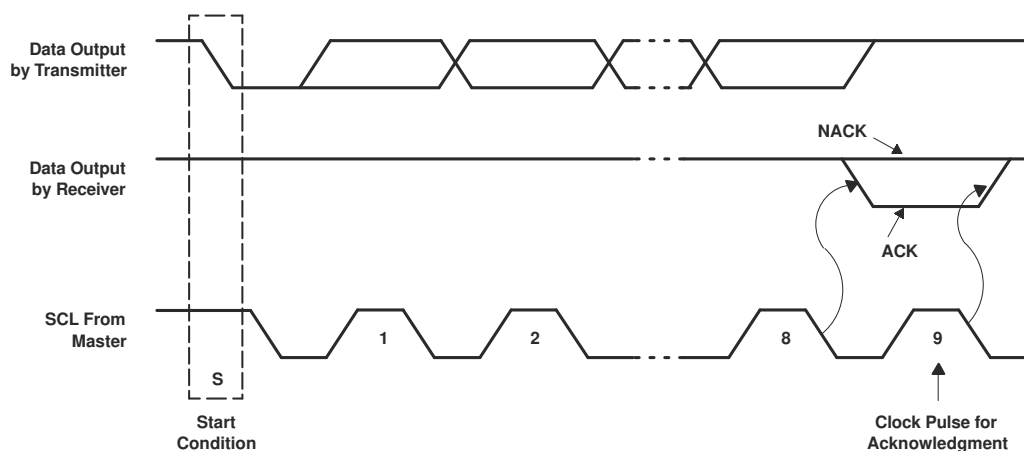
A master receiver will signal an end of data to the slave transmitter by not generating an acknowledge (NACK) after the last byte has been clocked out of the slave. This is done by the master receiver by holding the SDA line high. In this event, the transmitter must release the data line to enable the master to generate a stop condition.



**Figure 9-3. Definition Of Start And Stop Conditions**



**Figure 9-4. Bit Transfer**



**Figure 9-5. Acknowledgment On I²C Bus**

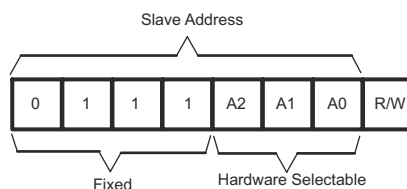
### 9.3.2 Register Map

**Table 9-1. Interface Definition**

| BYTE              | BIT     |    |    |    |    |    |    |              |
|-------------------|---------|----|----|----|----|----|----|--------------|
|                   | 7 (MSB) | 6  | 5  | 4  | 3  | 2  | 1  | 0 (LSB)      |
| I²C slave address | L       | H  | H  | H  | A2 | A1 | A0 | R/ $\bar{W}$ |
| Px I/O data bus   | P7      | P6 | P5 | P4 | P3 | P2 | P1 | P0           |

#### 9.3.2.1 Device Address

Figure 9-6 shows the address byte of the PCA9534A.



**Figure 9-6. Pca9534a Address**

**Table 9-2. Address Reference**

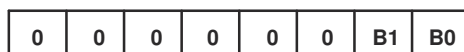
| INPUTS |    |    | I <sup>2</sup> C BUS SLAVE ADDRESS |
|--------|----|----|------------------------------------|
| A2     | A1 | A0 |                                    |
| L      | L  | L  | 56 (decimal), 38 (hexadecimal)     |
| L      | L  | H  | 57 (decimal), 39 (hexadecimal)     |
| L      | H  | L  | 58 (decimal), 3A (hexadecimal)     |
| L      | H  | H  | 59 (decimal), 3B (hexadecimal)     |
| H      | L  | L  | 60 (decimal), 3C (hexadecimal)     |
| H      | L  | H  | 61 (decimal), 3D (hexadecimal)     |
| H      | H  | L  | 62 (decimal), 3E (hexadecimal)     |
| H      | H  | H  | 63 (decimal), 3F (hexadecimal)     |

The last bit of the slave address defines the operation (read or write) to be performed. When it is high (1), a read is selected, while a low (0) selects a write operation.

### 9.3.2.2 Control Register And Command Byte

Following the successful acknowledgment of the address byte, the bus master sends a command byte which is stored in the control register in the PCA9534A. Two bits of this command byte state the operation (read or write) and the internal register (input, output, polarity inversion or configuration) that will be affected. This register can be written or read through the I<sup>2</sup>C bus. The command byte is sent only during a write transmission.

Once a command byte has been sent, the register that was addressed continues to be accessed by reads until a new command byte has been sent.



**Figure 9-7. Control Register Bits**

**Table 9-3. Command Byte**

| CONTROL REGISTER BITS |    | COMMAND BYTE (HEX) | REGISTER           | PROTOCOL        | POWER-UP DEFAULT |
|-----------------------|----|--------------------|--------------------|-----------------|------------------|
| B1                    | B0 |                    |                    |                 |                  |
| 0                     | 0  | 0x00               | Input Port         | Read byte       | xxxx xxxx        |
| 0                     | 1  | 0x01               | Output Port        | Read/write byte | 1111 1111        |
| 1                     | 0  | 0x02               | Polarity Inversion | Read/write byte | 0000 0000        |
| 1                     | 1  | 0x03               | Configuration      | Read/write byte | 1111 1111        |

### 9.3.2.3 Register Descriptions

The input port register (register 0) reflects the incoming logic levels of the pins, regardless of whether the pin is defined as an input or an output by the configuration register. It only acts on read operation. Writes to these registers have no effect. The default value, X, is determined by the externally applied logic level.

Before a read operation, a write transmission is sent with the command byte to let the I<sup>2</sup>C device know that the input port register will be accessed next.

**Table 9-4. Register 0 (Input Port Register)**

| BIT     | I7 | I6 | I5 | I4 | I3 | I2 | I1 | I0 |
|---------|----|----|----|----|----|----|----|----|
| DEFAULT | X  | X  | X  | X  | X  | X  | X  | X  |

The output port register (register 1) shows the outgoing logic levels of the pins defined as outputs by the configuration register. Bit values in this register have no effect on pins defined as inputs. In turn, reads from this register reflect the value that is in the flip-flop controlling the output selection, not the actual pin value.

**Table 9-5. Register 1 (Output Port Register)**

| BIT     | O7 | O6 | O5 | O4 | O3 | O2 | O1 | O0 |
|---------|----|----|----|----|----|----|----|----|
| DEFAULT | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  |

The polarity inversion register (register 2) allows polarity inversion of pins defined as inputs by the configuration register. If a bit in this register is set (written with 1), the corresponding port pin polarity is inverted. If a bit in this register is cleared (written with a 0), the corresponding port pin original polarity is retained.

**Table 9-6. Register 2 (Polarity Inversion Register)**

| BIT     | N7 | N6 | N5 | N4 | N3 | N2 | N1 | N0 |
|---------|----|----|----|----|----|----|----|----|
| DEFAULT | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

The configuration register (register 3) configures the directions of the I/O pins. If a bit in this register is set to 1, the corresponding port pin is enabled as an input with high-impedance output driver. If a bit in this register is cleared to 0, the corresponding port pin is enabled as an output.

**Table 9-7. Register 3 (Configuration Register)**

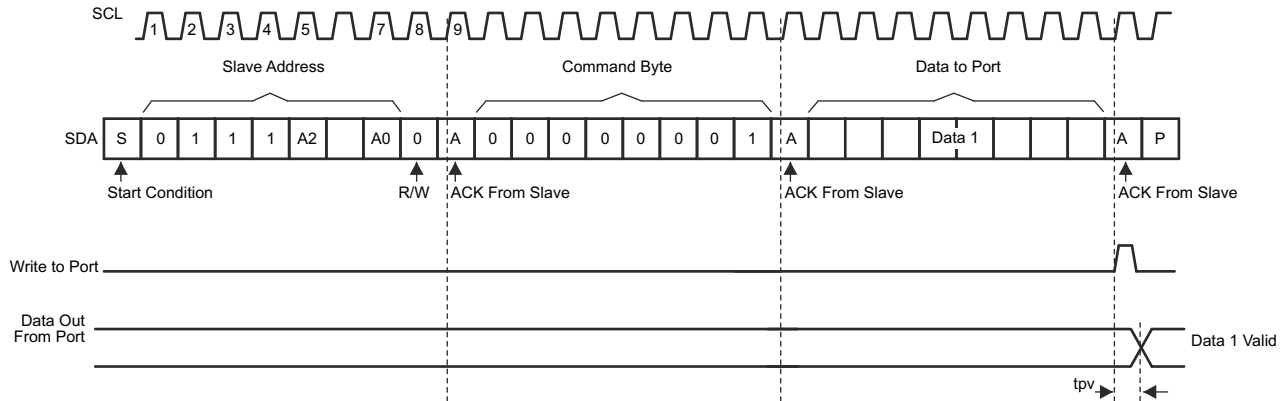
| BIT     | C7 | C6 | C5 | C4 | C3 | C2 | C1 | C0 |
|---------|----|----|----|----|----|----|----|----|
| DEFAULT | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  |

### 9.3.2.4 Bus Transactions

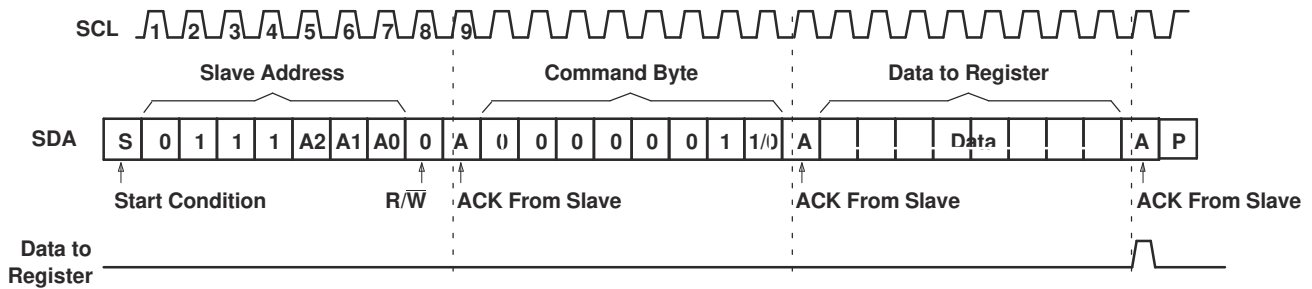
Data is exchanged between the master and PCA9534A through write and read commands.

#### 9.3.2.4.1 Writes

Data is transmitted to the PCA9534A by sending the device address and setting the least-significant bit to a logic 0 (see [Figure 9-6](#) for device address). The command byte is sent after the address and determines which register receives the data that follows the command byte (see [Figure 9-8](#) and [Figure 9-9](#)). There is no limitation on the number of data bytes sent in one write transmission.



**Figure 9-8. Write To Output Port Register**



**Figure 9-9. Write To Configuration Or Polarity Inversion Registers**

### 9.3.2.4.2 Reads

The bus master first must send the PCA9534A address with the least-significant bit set to a logic 0 (see [Figure 9-6](#) for device address). The command byte is sent after the address and determines which register is accessed. After a restart, the device address is sent again but, this time, the least-significant bit is set to a logic 1. Data from the register defined by the command byte then is sent by the PCA9534A (see [Figure 9-10](#) and [Figure 9-11](#)). After a restart, the value of the register defined by the command byte matches the register being accessed when the restart occurred. Data is clocked into the register on the rising edge of the ACK clock pulse. There is no limitation on the number of data bytes received in one read transmission, but when the final byte is received, the bus master must not acknowledge the data.

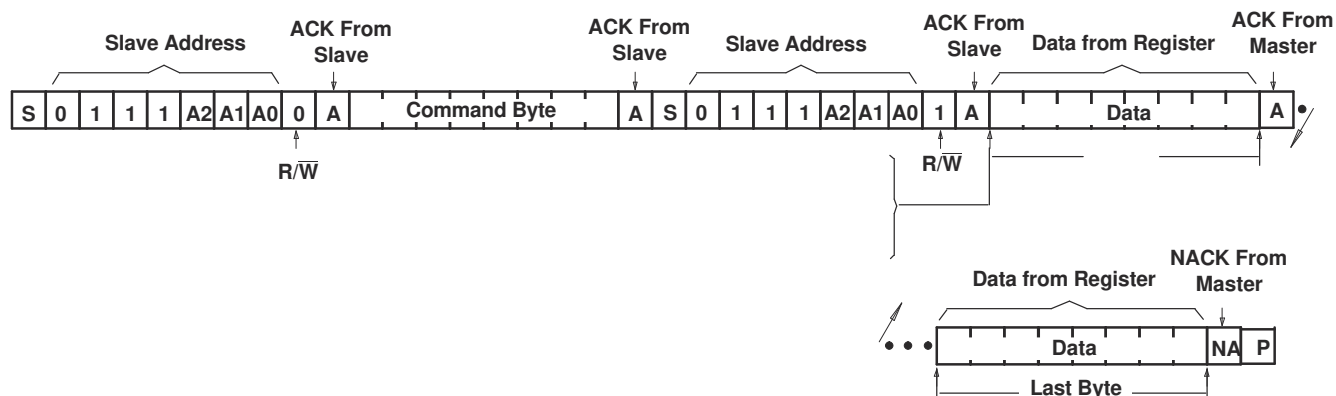
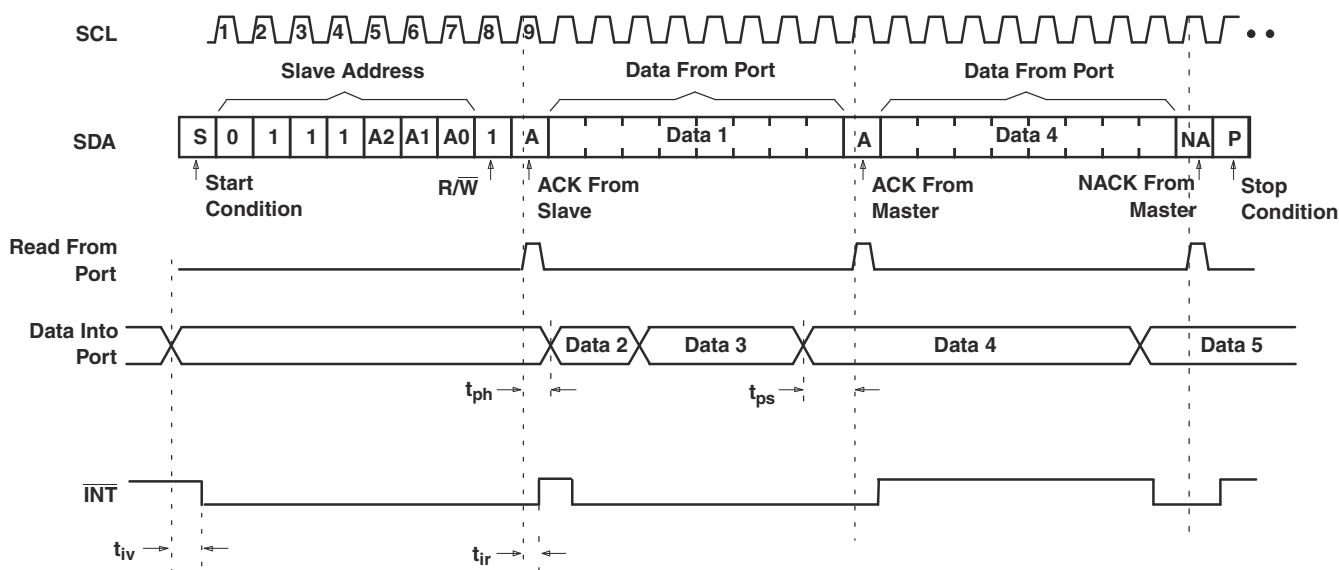


Figure 9-10. Read From Register



- A. This figure assumes that the command byte has previously been programmed with 00h.
- B. Transfer of data can be stopped at any moment by a stop condition.
- C. This figure eliminates the command byte transfer, a restart and slave address call between the initial slave address call and the actual data transfer from the P Port. See [Figure 9-10](#) for these details.

Figure 9-11. Read Input Port Register

## 10 Application Information Disclaimer

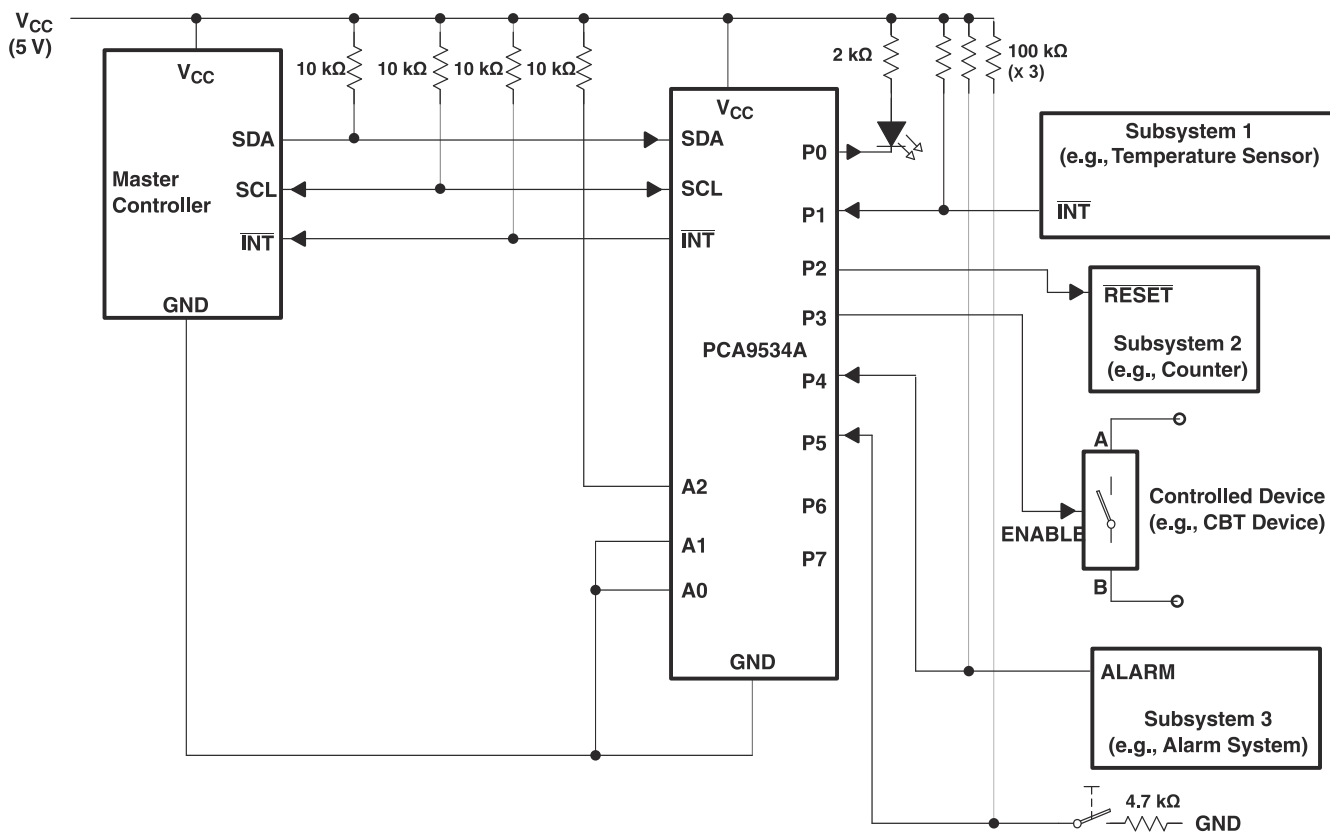
### Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

## 10.1 Application Information

### 10.1.1 Typical Application

Figure 10-1 shows an application in which the PCA9534A can be used.



- A. Device address is configured as 0111100 for this example.
- B. P0, P2, and P3 are configured as outputs.
- C. P1, P4, and P5 are configured as inputs.
- D. P6 and P7 are not used and must be configured as outputs.

**Figure 10-1. Typical Application**

### 10.1.1.1 Design Requirements

#### 10.1.1.1.1 Minimizing $I_{CC}$ When The I/O Controls Leds

When the I/Os are used to control LEDs, they normally are connected to  $V_{CC}$  through a resistor as shown in Figure 10-1. Because the LED acts as a diode, when the LED is off, the I/O  $V_{IN}$  is about 1.2 V less than  $V_{CC}$ . The supply current,  $I_{CC}$ , increases as  $V_{IN}$  becomes lower than  $V_{CC}$  and is specified as  $\Delta I_{CC}$  in *Electrical Characteristics*.

For battery-powered applications, it is essential that the voltage of the I/O pins is greater than or equal to  $V_{CC}$  when the LED is off to minimize current consumption. Figure 10-2 shows a high-value resistor in parallel with the LED. Figure 10-3 shows  $V_{CC}$  less than the LED supply voltage by at least 1.2 V. Both of these methods maintain the I/O  $V_{IN}$  at or above  $V_{CC}$  and prevents additional supply-current consumption when the LED is off.

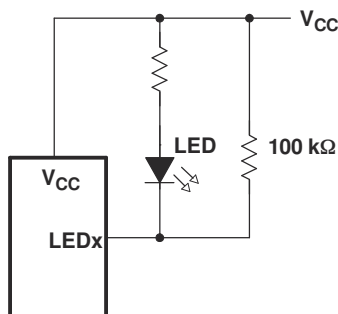


Figure 10-2. High-Value Resistor In Parallel With The Led

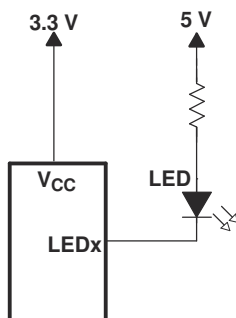


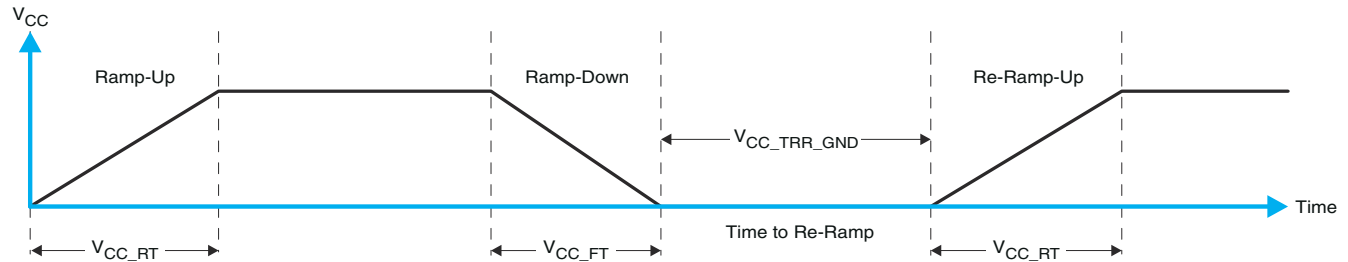
Figure 10-3. Device Supplied By A Lower Voltage

## 11 Power Supply Recommendations

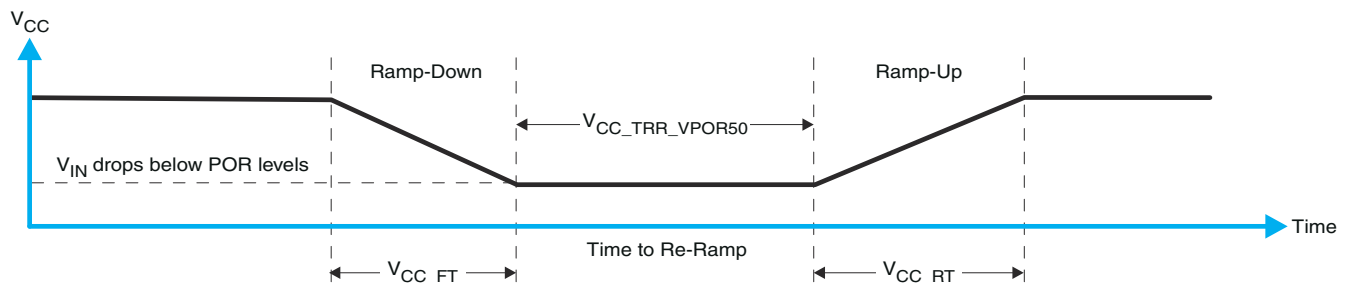
### 11.1 Power-On Reset Requirements

In the event of a glitch or data corruption, PCA9534A can be reset to its default conditions by using the power-on reset feature. Power-on reset requires that the device go through a power cycle to be completely reset. This reset also happens when the device is powered on for the first time in an application.

The two types of power-on reset are shown in [Figure 11-1](#) and [Figure 11-2](#).



**Figure 11-1.  $V_{CC}$  Is Lowered Below 0.2 V Or 0 V And Then Ramped Up To  $V_{CC}$**



**Figure 11-2.  $V_{CC}$  Is Lowered Below The Por Threshold, Then Ramped Back Up To  $V_{CC}$**

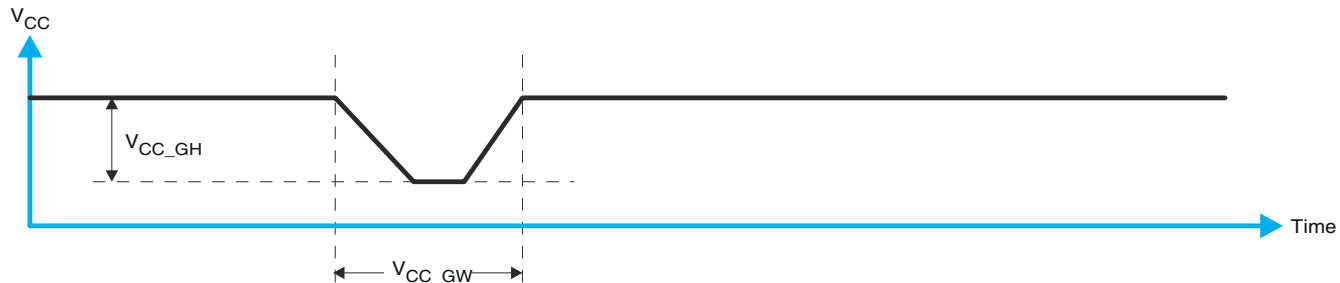
[Table 11-1](#) specifies the performance of the power-on reset feature for PCA9534A for both types of power-on reset.

**Table 11-1. Recommended Supply Sequencing And Ramp Rates <sup>(1)</sup>**

| PARAMETER            |                                                                                                               |                                 | MIN   | TYP | MAX   | UNIT    |
|----------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------|-------|-----|-------|---------|
| $V_{CC\_FT}$         | Fall rate                                                                                                     | See <a href="#">Figure 11-1</a> | 1     |     | 100   | ms      |
| $V_{CC\_RT}$         | Rise rate                                                                                                     | See <a href="#">Figure 11-1</a> | 0.01  |     | 100   | ms      |
| $V_{CC\_TRR\_GND}$   | Time to re-ramp (when $V_{CC}$ drops to GND)                                                                  | See <a href="#">Figure 11-1</a> | 0.001 |     |       | ms      |
| $V_{CC\_TRR\_POR50}$ | Time to re-ramp (when $V_{CC}$ drops to $V_{POR\_MIN} - 50$ mV)                                               | See <a href="#">Figure 11-2</a> | 0.001 |     |       | ms      |
| $V_{CC\_GH}$         | Level that $V_{CCP}$ can glitch down to, but not cause a functional disruption when $V_{CCX\_GW} = 1$ $\mu$ s | See <a href="#">Figure 11-3</a> |       |     | 1.2   | V       |
| $V_{CC\_GW}$         | Glitch width that will not cause a functional disruption when $V_{CCX\_GH} = 0.5 \times V_{CCX}$              | See <a href="#">Figure 11-3</a> |       |     |       | $\mu$ s |
| $V_{PORF}$           | Voltage trip point of POR on falling $V_{CC}$                                                                 |                                 | 0.767 |     | 1.144 | V       |
| $V_{PORR}$           | Voltage trip point of POR on rising $V_{CC}$                                                                  |                                 | 1.033 |     | 1.428 | V       |

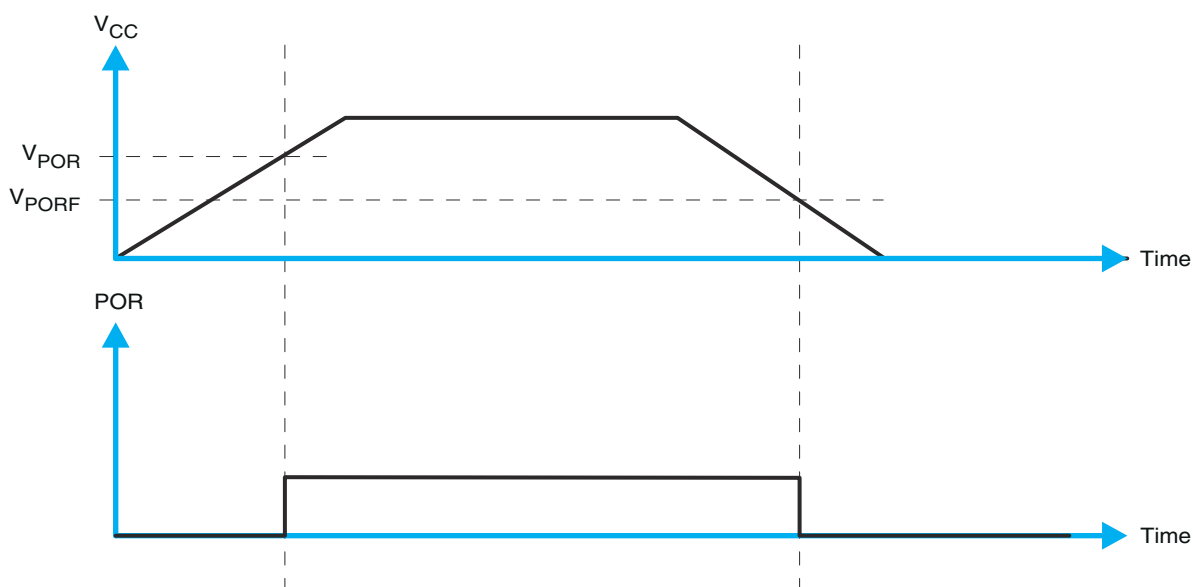
(1)  $T_A = -40^\circ\text{C}$  to  $85^\circ\text{C}$  (unless otherwise noted)

Glitches in the power supply can also affect the power-on reset performance of this device. The glitch width ( $V_{CC\_GW}$ ) and height ( $V_{CC\_GH}$ ) are dependent on each other. The bypass capacitance, source impedance, and the device impedance are factors that affect power-on reset performance. [Figure 11-3](#) and [Table 11-1](#) provide more information on how to measure these specifications.



**Figure 11-3. Glitch Width And Glitch Height**

$V_{POR}$  is critical to the power-on reset.  $V_{POR}$  is the voltage level at which the reset condition is released and all the registers and the I<sup>2</sup>C/SMBus state machine are initialized to their default states. The value of  $V_{POR}$  differs based on the  $V_{CC}$  being lowered to or from 0. [Figure 11-4](#) and [Table 11-1](#) provide more details on this specification.



**Figure 11-4.  $V_{POR}$**

## 12 Device and Documentation Support

### 12.1 Trademarks

All trademarks are the property of their respective owners.

### 12.2 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

### 12.3 Glossary

**TI Glossary** This glossary lists and explains terms, acronyms, and definitions.

## 13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

## PACKAGING INFORMATION

| Orderable part number        | Status<br>(1) | Material type<br>(2) | Package   Pins   | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6) |
|------------------------------|---------------|----------------------|------------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|---------------------|
| <a href="#">PCA9534ADB</a>   | Active        | Production           | SSOP (DB)   16   | 80   TUBE             | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | PD534A              |
| PCA9534ADB.A                 | Active        | Production           | SSOP (DB)   16   | 80   TUBE             | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | PD534A              |
| <a href="#">PCA9534ADBR</a>  | Active        | Production           | SSOP (DB)   16   | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | PD534A              |
| PCA9534ADBR.A                | Active        | Production           | SSOP (DB)   16   | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | PD534A              |
| <a href="#">PCA9534ADGVR</a> | Active        | Production           | TVSOP (DGV)   16 | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | PD534A              |
| PCA9534ADGVR.A               | Active        | Production           | TVSOP (DGV)   16 | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | PD534A              |
| <a href="#">PCA9534ADW</a>   | Obsolete      | Production           | SOIC (DW)   16   | -                     | -           | Call TI                              | Call TI                           | -40 to 85    | PCA9534A            |
| <a href="#">PCA9534ADWR</a>  | Obsolete      | Production           | SOIC (DW)   16   | -                     | -           | Call TI                              | Call TI                           | -40 to 85    | PCA9534A            |
| <a href="#">PCA9534APWR</a>  | Active        | Production           | TSSOP (PW)   16  | 2000   LARGE T&R      | Yes         | NIPDAU   SN                          | Level-1-260C-UNLIM                | -40 to 85    | PD534A              |
| PCA9534APWR.A                | Active        | Production           | TSSOP (PW)   16  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | PD534A              |
| PCA9534APWR.B                | Active        | Production           | TSSOP (PW)   16  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | PD534A              |
| PCA9534APWRG4                | Active        | Production           | TSSOP (PW)   16  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | PD534A              |
| PCA9534APWRG4.A              | Active        | Production           | TSSOP (PW)   16  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | PD534A              |
| PCA9534APWRG4.B              | Active        | Production           | TSSOP (PW)   16  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | PD534A              |
| <a href="#">PCA9534ARGTR</a> | Active        | Production           | VQFN (RGT)   16  | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 85    | ZVJ                 |
| PCA9534ARGTR.A               | Active        | Production           | VQFN (RGT)   16  | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 85    | ZVJ                 |
| PCA9534ARGTR.B               | Active        | Production           | VQFN (RGT)   16  | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 85    | ZVJ                 |
| PCA9534ARGTRG4               | Active        | Production           | VQFN (RGT)   16  | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 85    | ZVJ                 |
| <a href="#">PCA9534ARGVR</a> | Active        | Production           | VQFN (RGV)   16  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 85    | PD534A              |
| PCA9534ARGVR.A               | Active        | Production           | VQFN (RGV)   16  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 85    | PD534A              |

<sup>(1)</sup> **Status:** For more details on status, see our [product life cycle](#).

<sup>(2)</sup> **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

<sup>(3)</sup> **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

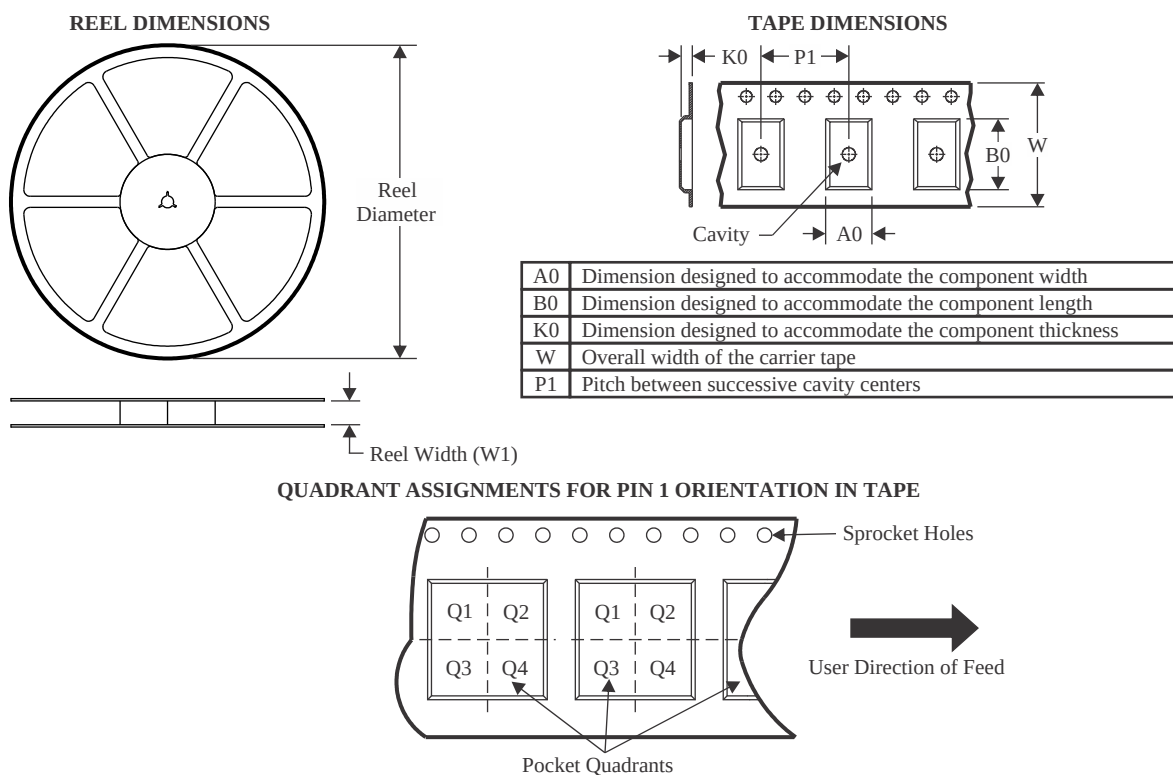
(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

**Important Information and Disclaimer:** The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

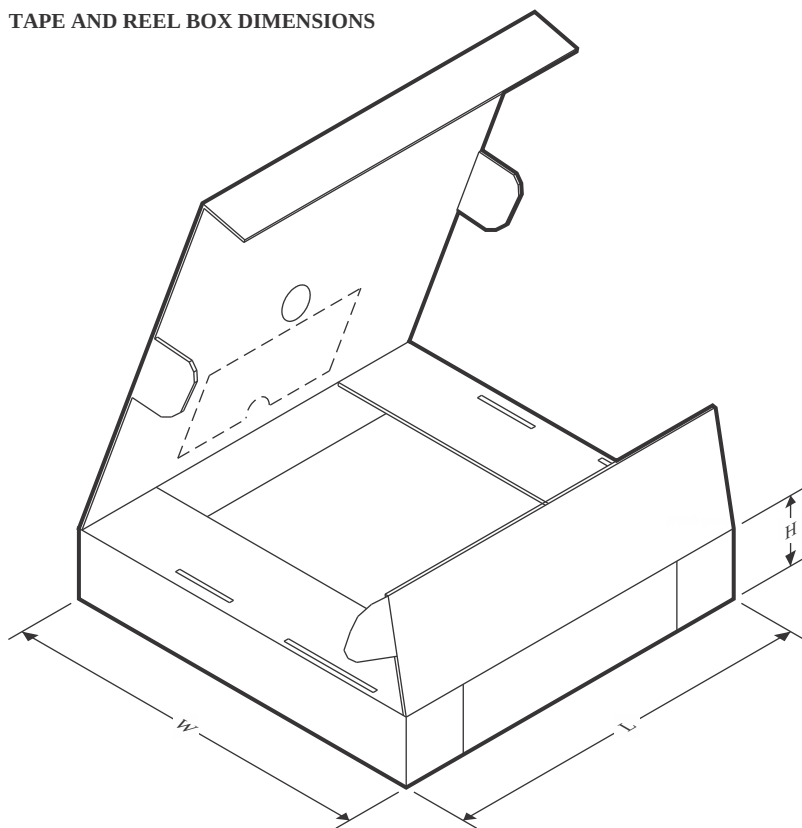
## TAPE AND REEL INFORMATION



\*All dimensions are nominal

| Device        | Package Type | Package Drawing | Pins | SPQ  | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---------------|--------------|-----------------|------|------|--------------------|--------------------|---------|---------|---------|---------|--------|---------------|
| PCA9534ADBR   | SSOP         | DB              | 16   | 2000 | 330.0              | 16.4               | 8.35    | 6.6     | 2.4     | 12.0    | 16.0   | Q1            |
| PCA9534ADGVR  | TVSOP        | DGV             | 16   | 2000 | 330.0              | 12.4               | 6.8     | 4.0     | 1.6     | 8.0     | 12.0   | Q1            |
| PCA9534APWR   | TSSOP        | PW              | 16   | 2000 | 330.0              | 12.4               | 6.9     | 5.6     | 1.6     | 8.0     | 12.0   | Q1            |
| PCA9534APWR   | TSSOP        | PW              | 16   | 2000 | 330.0              | 12.4               | 6.9     | 5.6     | 1.6     | 8.0     | 12.0   | Q1            |
| PCA9534APWR   | TSSOP        | PW              | 16   | 2000 | 330.0              | 12.4               | 6.9     | 5.6     | 1.6     | 8.0     | 12.0   | Q1            |
| PCA9534APWRG4 | TSSOP        | PW              | 16   | 2000 | 330.0              | 12.4               | 6.9     | 5.6     | 1.6     | 8.0     | 12.0   | Q1            |
| PCA9534APWRG4 | TSSOP        | PW              | 16   | 2000 | 330.0              | 12.4               | 6.9     | 5.6     | 1.6     | 8.0     | 12.0   | Q1            |
| PCA9534ARGTR  | VQFN         | RGT             | 16   | 3000 | 330.0              | 12.4               | 3.3     | 3.3     | 1.1     | 8.0     | 12.0   | Q2            |
| PCA9534ARGVR  | VQFN         | RGV             | 16   | 2500 | 330.0              | 12.4               | 4.25    | 4.25    | 1.15    | 8.0     | 12.0   | Q2            |

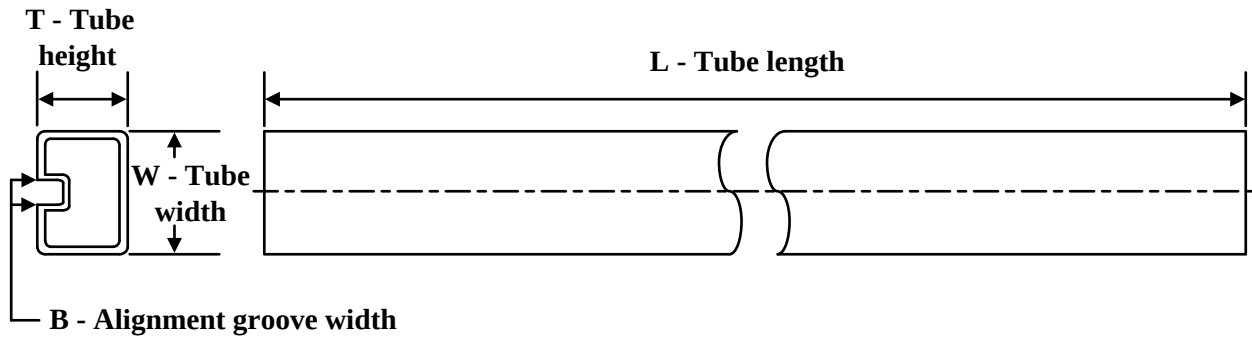
## TAPE AND REEL BOX DIMENSIONS



\*All dimensions are nominal

| Device        | Package Type | Package Drawing | Pins | SPQ  | Length (mm) | Width (mm) | Height (mm) |
|---------------|--------------|-----------------|------|------|-------------|------------|-------------|
| PCA9534ADBR   | SSOP         | DB              | 16   | 2000 | 353.0       | 353.0      | 32.0        |
| PCA9534ADGVR  | TVSOP        | DGV             | 16   | 2000 | 353.0       | 353.0      | 32.0        |
| PCA9534APWR   | TSSOP        | PW              | 16   | 2000 | 353.0       | 353.0      | 32.0        |
| PCA9534APWR   | TSSOP        | PW              | 16   | 2000 | 353.0       | 353.0      | 32.0        |
| PCA9534APWR   | TSSOP        | PW              | 16   | 2000 | 356.0       | 356.0      | 35.0        |
| PCA9534APWRG4 | TSSOP        | PW              | 16   | 2000 | 353.0       | 353.0      | 32.0        |
| PCA9534APWRG4 | TSSOP        | PW              | 16   | 2000 | 353.0       | 353.0      | 32.0        |
| PCA9534ARGTR  | VQFN         | RGT             | 16   | 3000 | 360.0       | 360.0      | 36.0        |
| PCA9534ARGVR  | VQFN         | RGV             | 16   | 2500 | 353.0       | 353.0      | 32.0        |

## TUBE



\*All dimensions are nominal

| Device       | Package Name | Package Type | Pins | SPQ | L (mm) | W (mm) | T (μm) | B (mm) |
|--------------|--------------|--------------|------|-----|--------|--------|--------|--------|
| PCA9534ADB   | DB           | SSOP         | 16   | 80  | 530    | 10.5   | 4000   | 4.1    |
| PCA9534ADB.A | DB           | SSOP         | 16   | 80  | 530    | 10.5   | 4000   | 4.1    |

## GENERIC PACKAGE VIEW

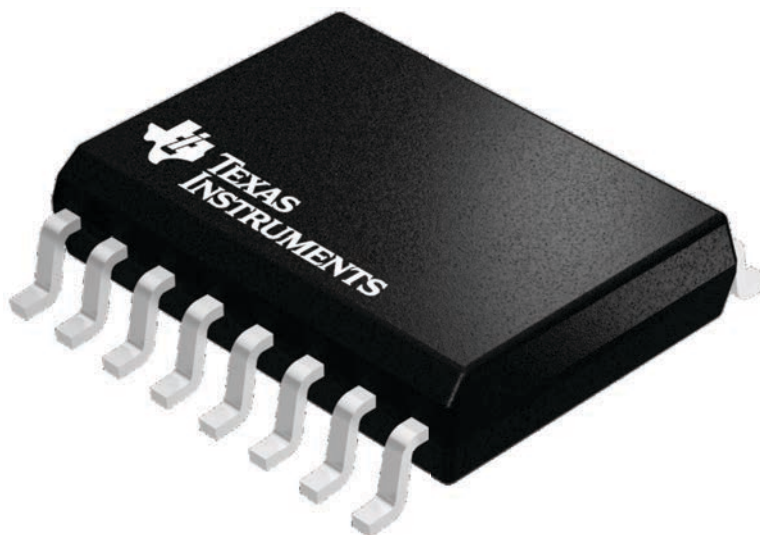
**DW 16**

**SOIC - 2.65 mm max height**

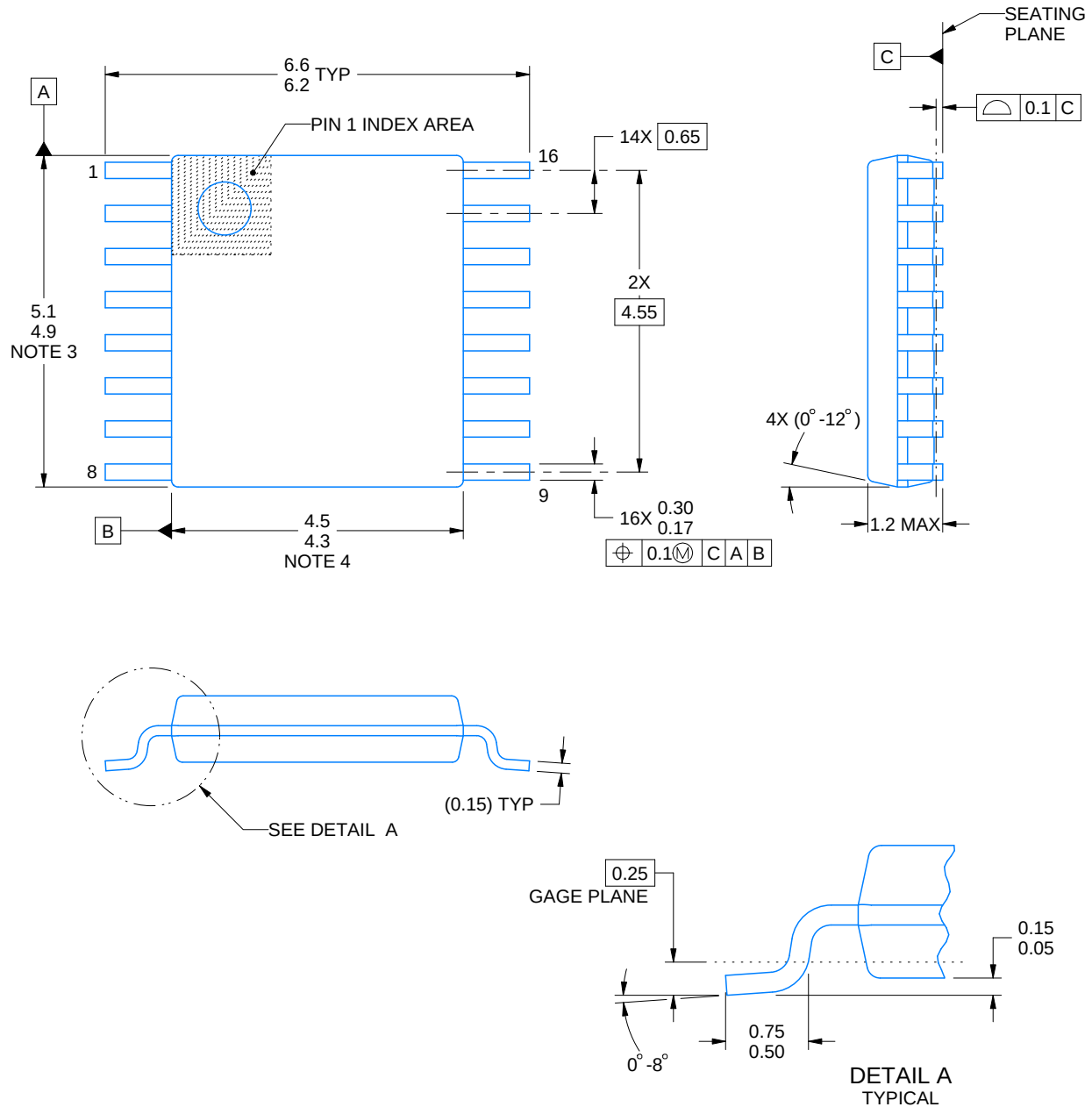
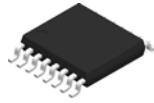
7.5 x 10.3, 1.27 mm pitch

SMALL OUTLINE INTEGRATED CIRCUIT

This image is a representation of the package family, actual package may vary.  
Refer to the product data sheet for package details.



4224780/A



4220204/B 12/2023

## NOTES:

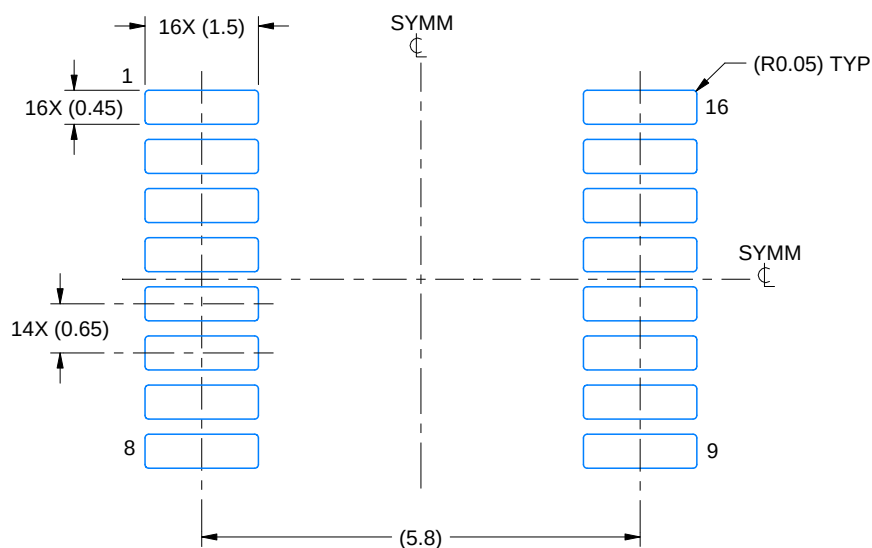
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.

# EXAMPLE BOARD LAYOUT

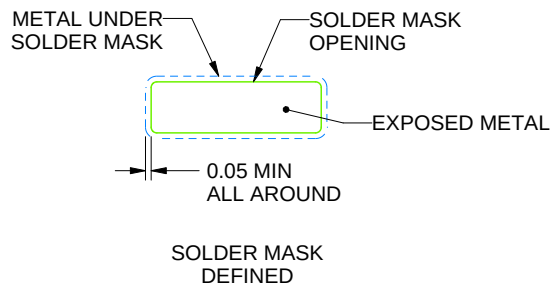
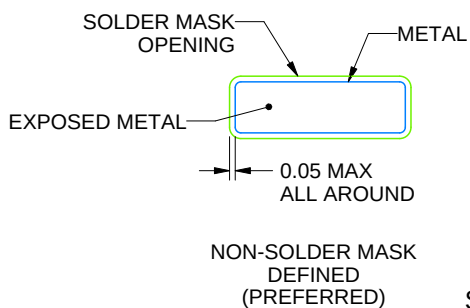
PW0016A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE: 10X



SOLDER MASK DETAILS

4220204/B 12/2023

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

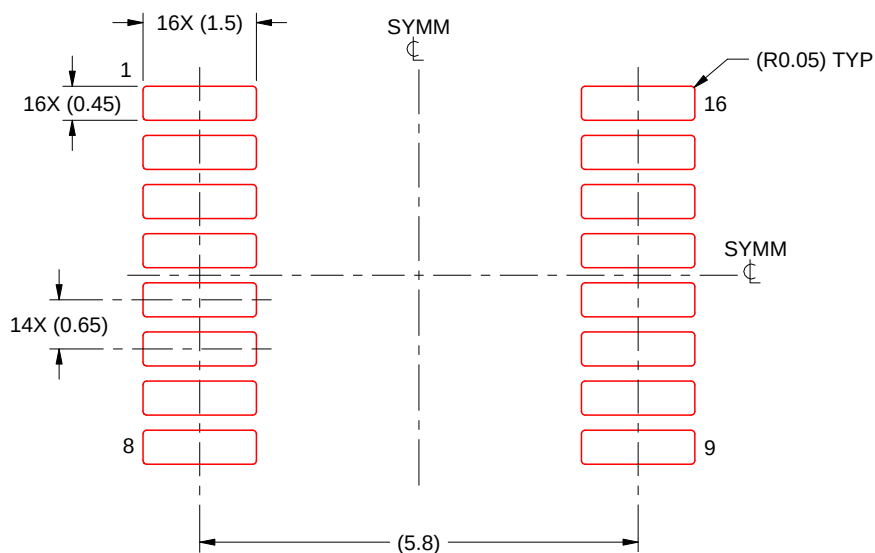
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

# EXAMPLE STENCIL DESIGN

PW0016A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE: 10X

4220204/B 12/2023

NOTES: (continued)

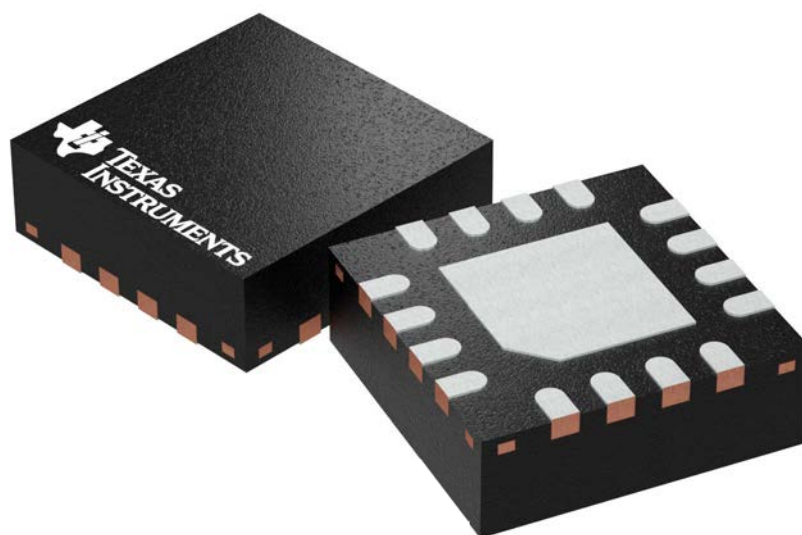
8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

**RGT 16**

**GENERIC PACKAGE VIEW**

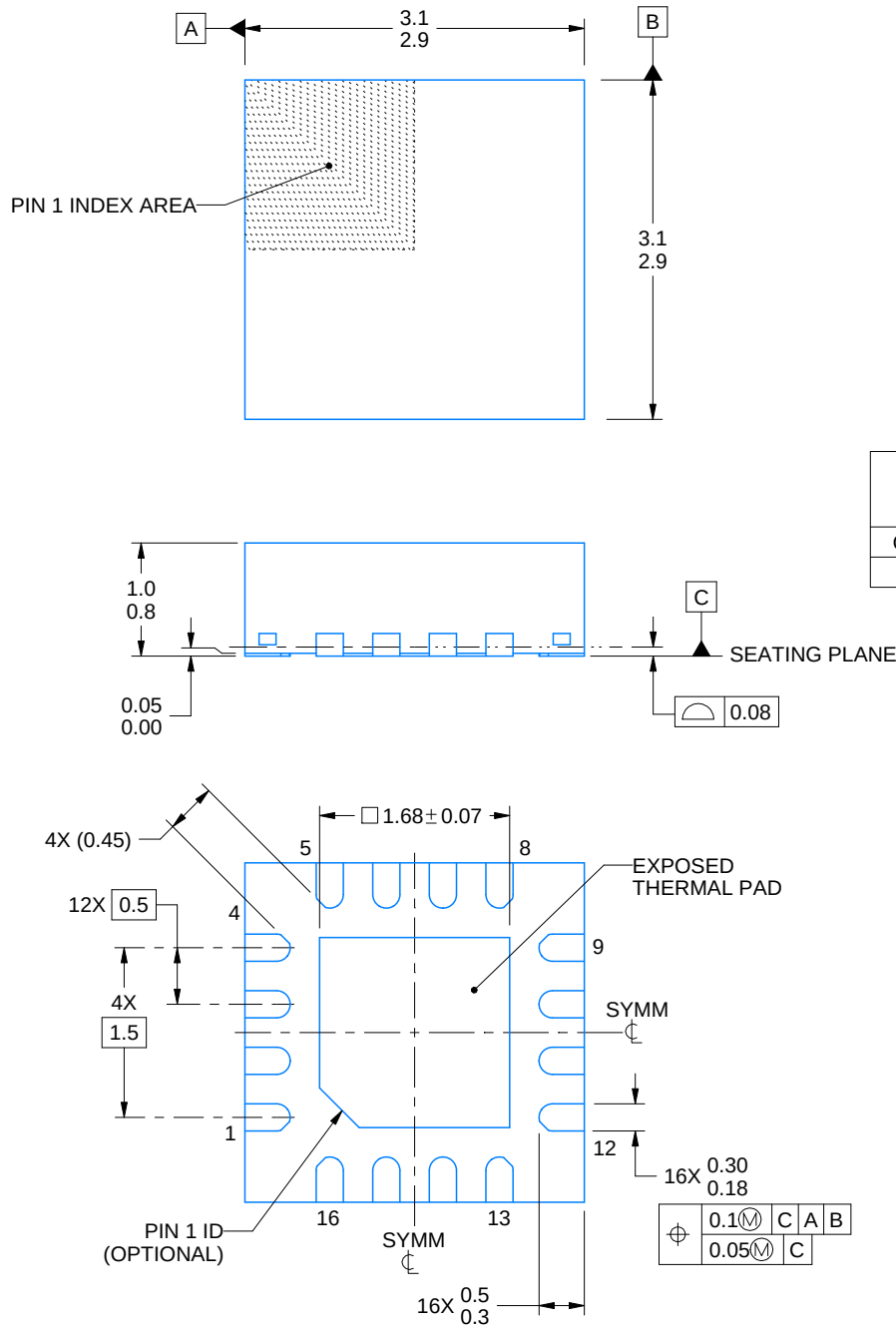
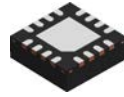
**VQFN - 1 mm max height**

PLASTIC QUAD FLATPACK - NO LEAD



Images above are just a representation of the package family, actual package may vary.  
Refer to the product data sheet for package details.

4203495/1



4222419/E 07/2025

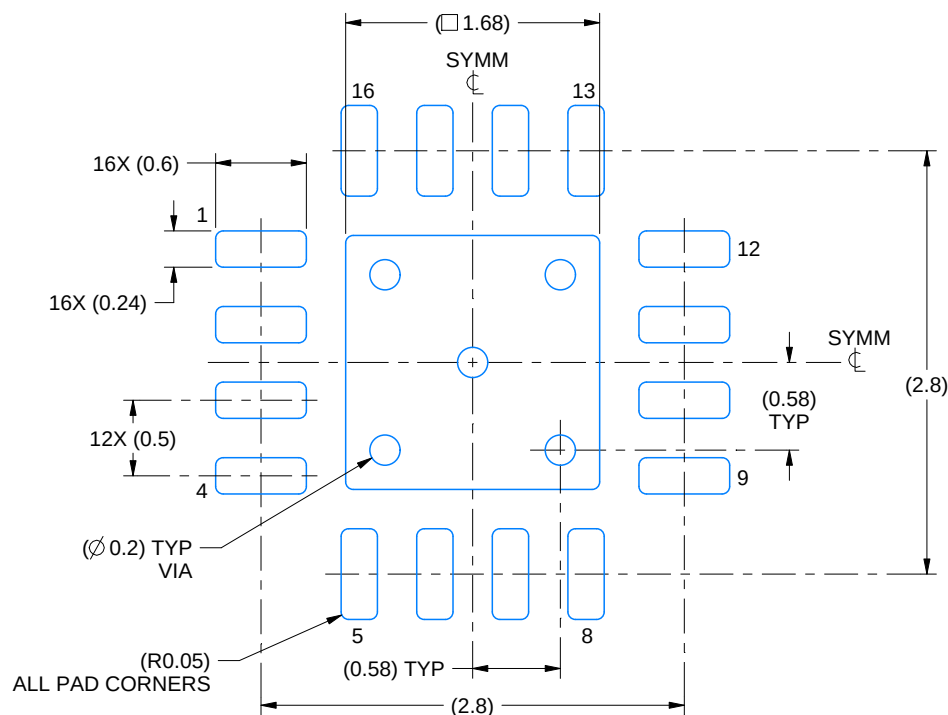
## NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. The package thermal pad must be soldered to the printed circuit board for thermal and mechanical performance.

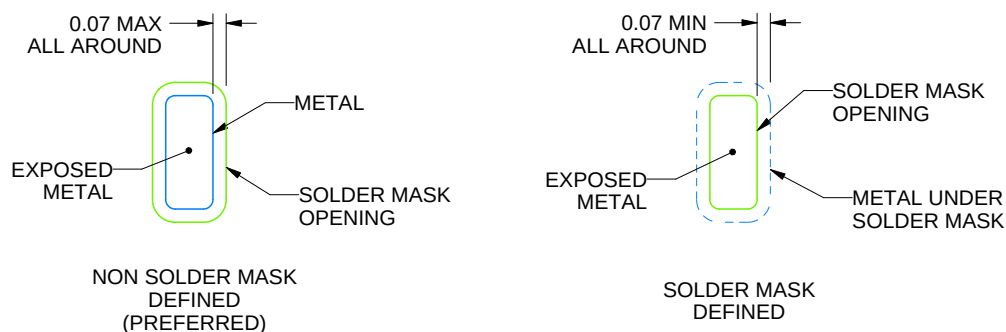
**RGT0016C**

### VQFN - 1 mm max height

PLASTIC QUAD FLATPACK - NO LEAD



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE:20X



## SOLDER MASK DETAILS

4222419/E 07/2025

NOTES: (continued)

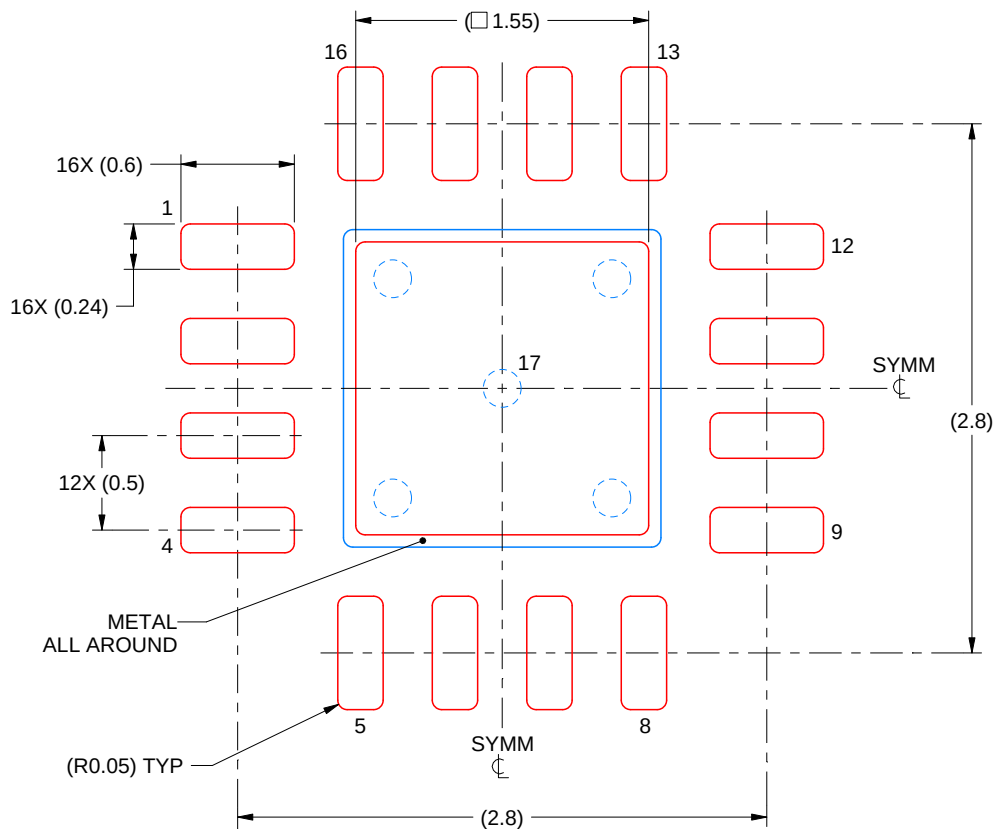
4. This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature number SLUA271 ([www.ti.com/lit/slua271](http://www.ti.com/lit/slua271)).
5. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.

# EXAMPLE STENCIL DESIGN

RGT0016C

VQFN - 1 mm max height

PLASTIC QUAD FLATPACK - NO LEAD



**SOLDER PASTE EXAMPLE**  
BASED ON 0.125 mm THICK STENCIL

EXPOSED PAD 17:  
85% PRINTED SOLDER COVERAGE BY AREA UNDER PACKAGE  
SCALE:25X

4222419/E 07/2025

NOTES: (continued)

6. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.

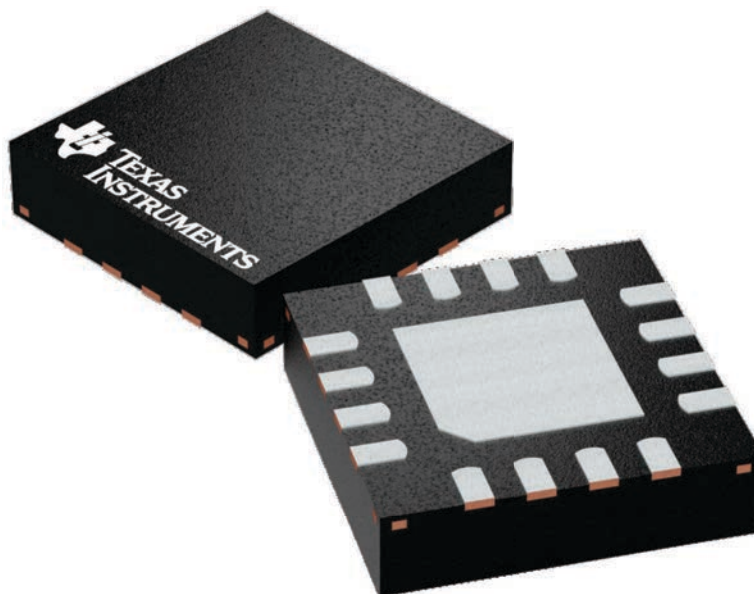
## GENERIC PACKAGE VIEW

**RGV 16**

**VQFN - 1 mm max height**

4 x 4, 0.65 mm pitch

PLASTIC QUAD FLATPACK - NO LEAD



Images above are just a representation of the package family, actual package may vary.  
Refer to the product data sheet for package details.

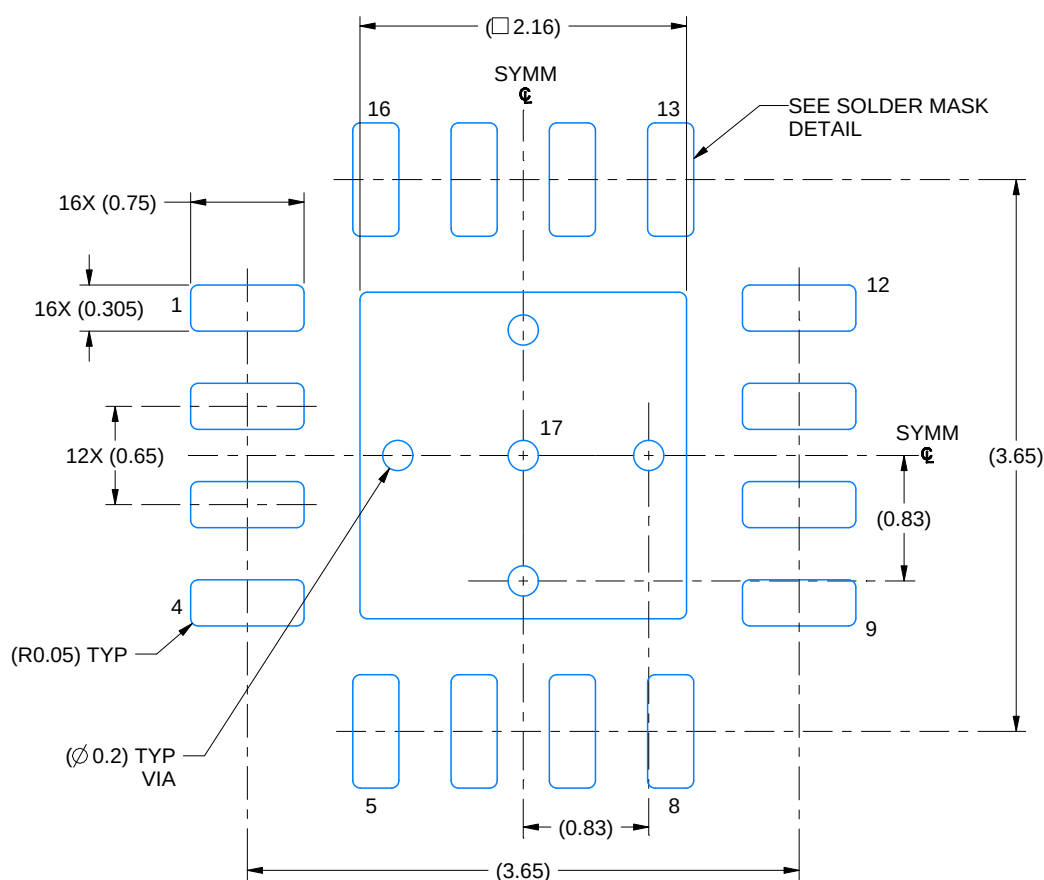
4224748/A

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. The package thermal pad must be soldered to the printed circuit board for thermal and mechanical performance.

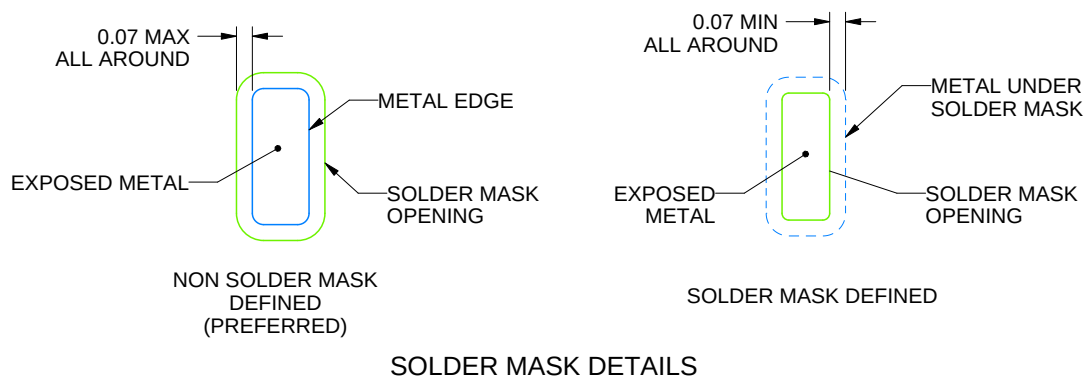
**RGV0016A**

### VQFN - 1 mm max height

PLASTIC QUAD FLATPACK - NO LEAD



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE: 20X



4219037/A 06/2019

NOTES: (continued)

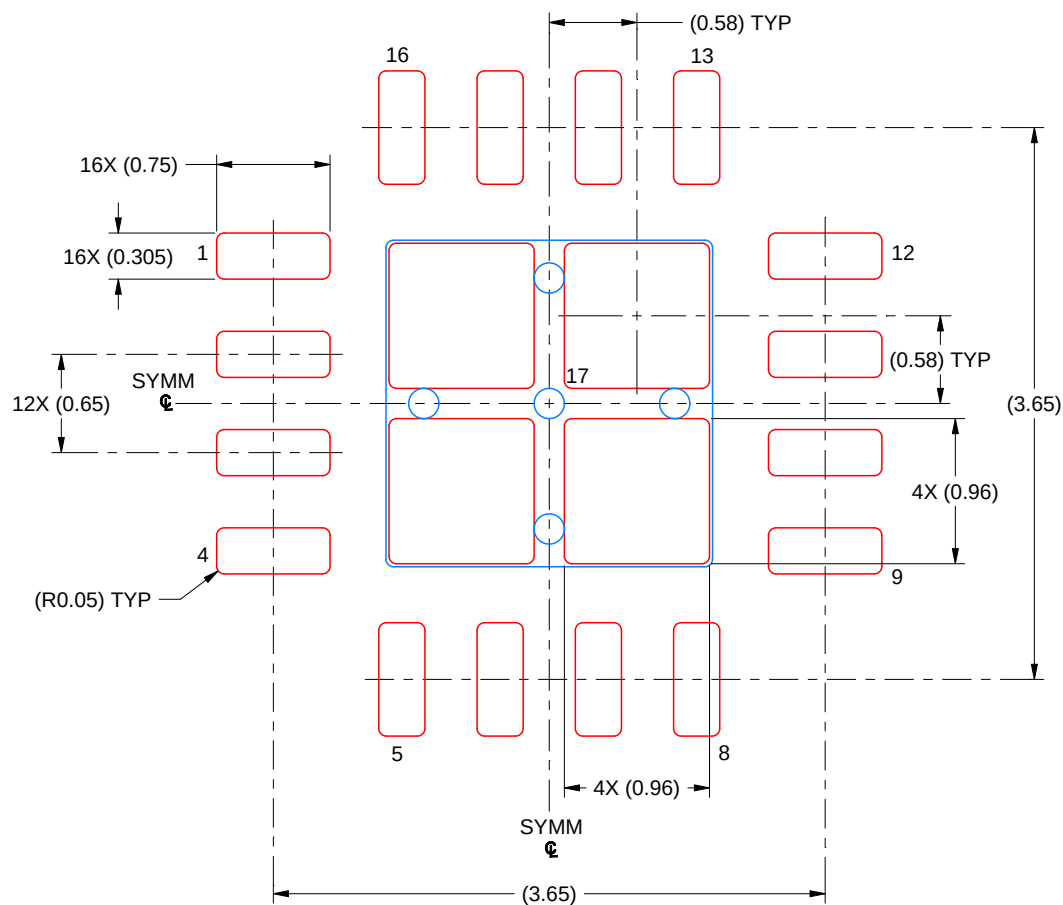
4. This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature number SLUA271 ([www.ti.com/lit/slua271](http://www.ti.com/lit/slua271)).
5. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.

# EXAMPLE STENCIL DESIGN

RGV0016A

VQFN - 1 mm max height

PLASTIC QUAD FLATPACK - NO LEAD



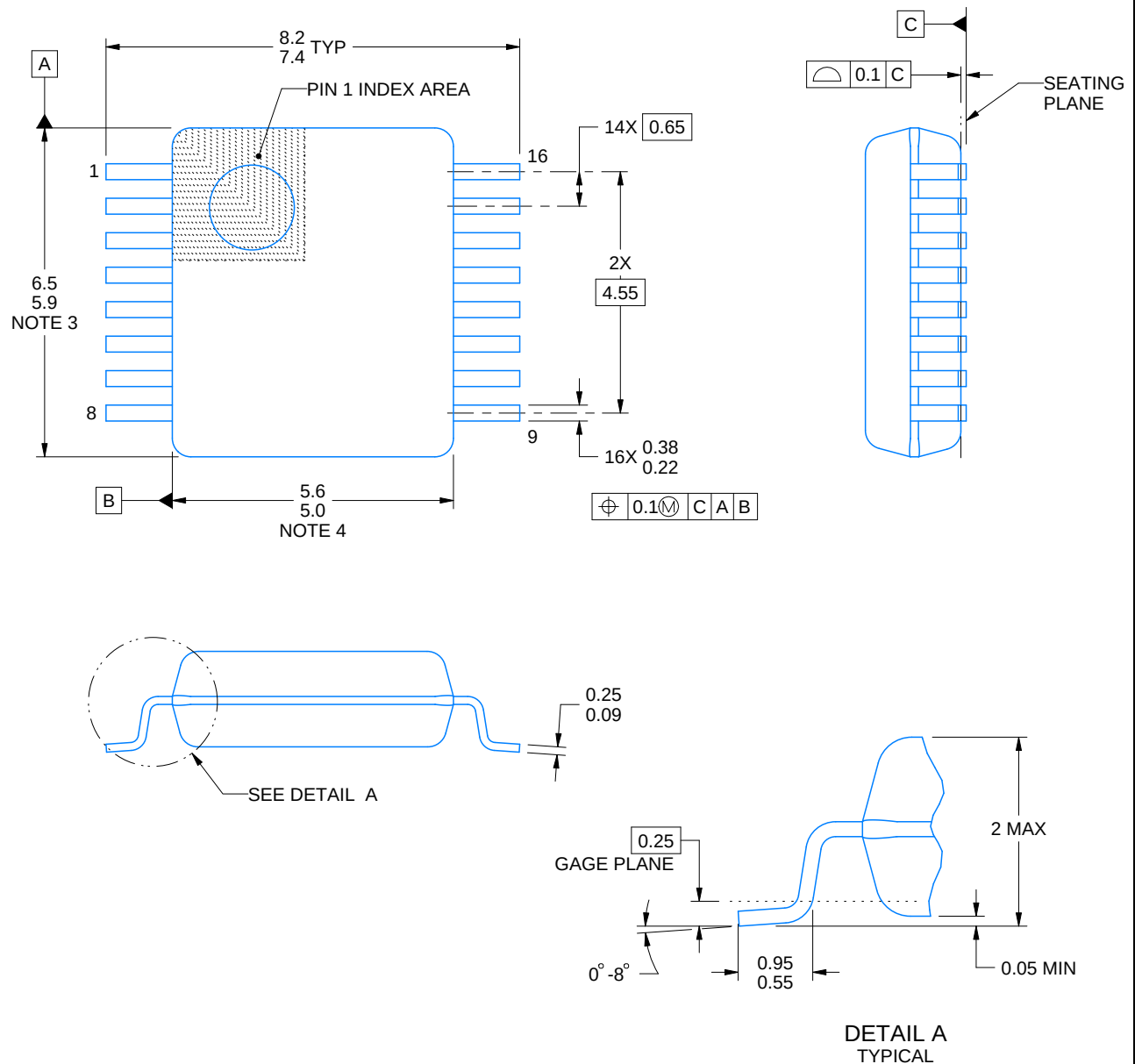
SOLDER PASTE EXAMPLE  
BASED ON 0.125 MM THICK STENCIL  
SCALE: 20X

EXPOSED PAD 17  
79% PRINTED SOLDER COVERAGE BY AREA UNDER PACKAGE

4219037/A 06/2019

NOTES: (continued)

6. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.



4220763/A 05/2022

## NOTES:

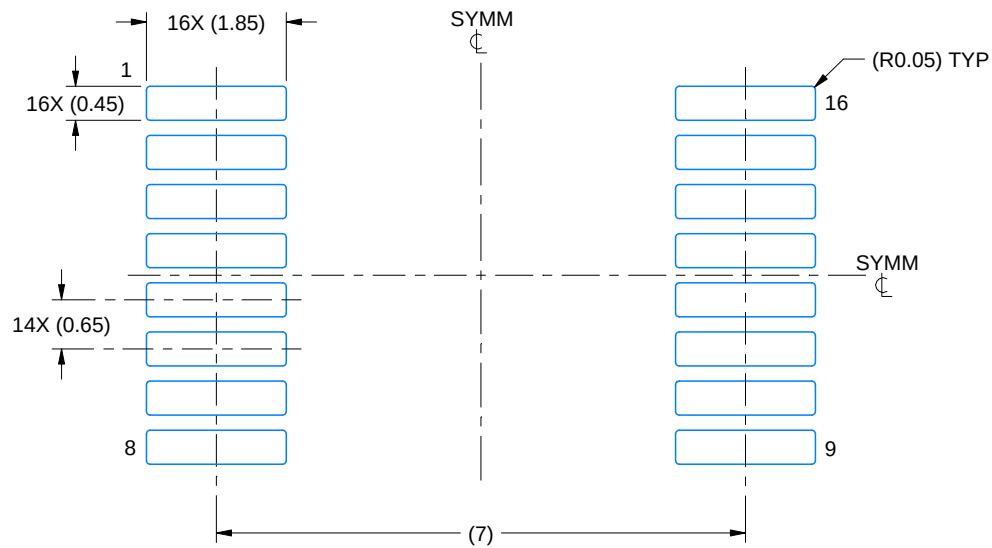
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. Reference JEDEC registration MO-150.

# EXAMPLE BOARD LAYOUT

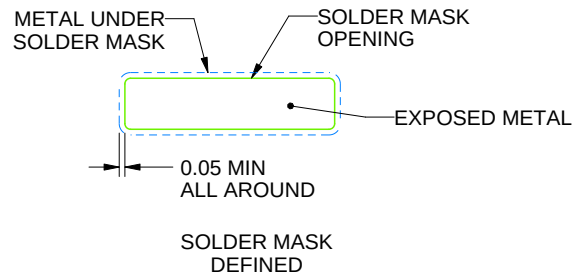
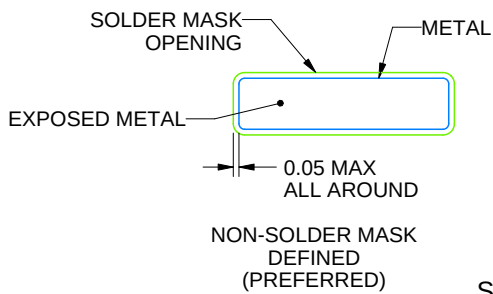
DB0016A

SSOP - 2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE: 10X



SOLDER MASK DETAILS

4220763/A 05/2022

NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.

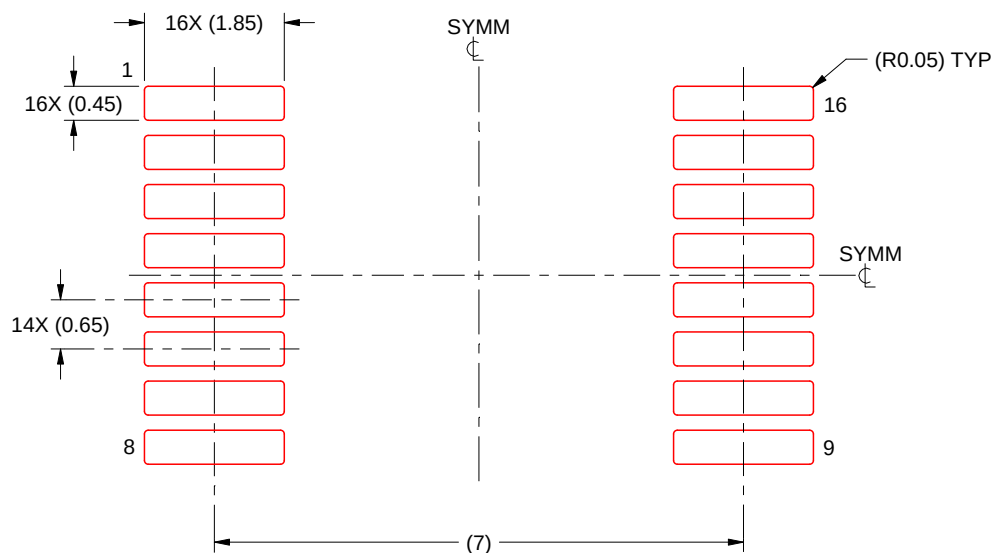
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

# EXAMPLE STENCIL DESIGN

DB0016A

SSOP - 2 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE: 10X

4220763/A 05/2022

NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

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